What Synthetic Data Buys and Costs: A Sim-to-Real Evaluation Case Study in Chess-Position Recognition

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Abstract

Perception models are judged on synthetic stress tests and on small real-photo sets. Both can mislead. We use chess-position recognition as a testbed, because ground truth is exact and one factor can be changed at a time. We describe a controlled render generator and a benchmark of 20,000 cell-images (18,000 unique renders). Its positions are disjoint, exactly and under the eight board symmetries, from 67 inventoried sources, including the named public chess-recognition datasets and a 6.1-million-puzzle snapshot. Five factors vary: camera elevation, piece geometry, board palette, framing, and the board colour convention (which corner is light).

Building the benchmark exposed three ways our earlier evaluation had misled us. First, a pipeline selected on a private leaderboard scored below its own single member on 16 of 18 controlled splits and on three real-photo or held-out cohorts; this is a descriptive comparison, since routing and member composition both differ. Second, two sets named as holdouts shared 1,446 and 1,128 image identifiers with the training mixes (2,398 distinct), a provenance failure. Third, a state-of-the-art recogniser produced zero exact boards on our renders, with per-square accuracy below a constant all-empty prediction. Two crops and a perspective warp recovered no exact board. A paired probe that flipped one renderer expression, the light/dark assignment of the a1 square, raised per-square accuracy from 54.4 to 85.2 and removed 67.7% of wrong squares, replicated across six piece families. On the 400-group development partition of the benchmark the same checkpoint scores 15.5 exact under the standard convention and 0.0 under ours.

A matched-position, fixed-update adaptation study then separates what a small budget of renders changes from what it does not. Low-rank adaptation on 500 positions (dark-only arm, standard convention) lifts the recogniser from 15.5 to 64.7 exact, and 8,000 positions to 79.7 (one seed each). Rendering each position under both conventions removes most of the convention gap (a1-light within 0.7–2.6 points of a1-dark), at a cost of 2.7 exact points on the standard convention at the 2,000-position budget (three seeds per arm, every both-convention seed below every dark-only seed) and single-seed differences of +2.10 and +0.95 at the other two. Every adapted checkpoint loses 11–23 boards on ChessReD, a preregistered retention gate that all ten runs fail. The lowest elevation band reaches only 14.5 exact after the largest budget. Generator, manifests, receipts, and the test-split hash list are prepared for release.

1 Introduction

A perception module is usually judged twice. Synthetic stress tests give free labels and controllable factors. Small sets of real photographs give expensive labels and thin coverage. Each judgement can mislead on its own. A synthetic test that shares a rendering lineage with the training data reports in-distribution accuracy while looking like a robustness test. A real-photo set that has been

scored during development becomes a development set. A pipeline chosen by its score on a small private leaderboard carries whatever that leaderboard rewarded, whether or not it generalises. We met all three while building a chess-position recogniser. This paper is the controlled follow-up.

Chess is a convenient testbed. Ground truth is exact: a placement FEN (Forsyth–Edwards Notation) names every one of 64 squares. Positions can be generated legally and checked for duplicates against named public pools. Renders are cheap, so one factor can be varied while everything else is held fixed. Public sets exist, both real photographs Masouris and van Gemert [2024], Seytre [2026b], Abeykoon et al. [2025] and renders Wölflein and Arandjelović [2021], Kojić [2024], and a strong public recogniser with released weights exists Seytre [2026a], so external baselines need no retraining. The domain is narrow. What we offer are evaluation practices that were cheap to apply here and that other sim-to-real pipelines can try, not a demonstrated transfer. The setting is the common one: a perception module trained on renders and deployed on a camera, as in robot manipulation, visual inspection and augmented reality.

Contributions.

1. A controlled render generator and a benchmark (Section 3) of 20,000 cell-images, 18,000 of them unique renders, with exact projected corners and five single-factor axes including the board colour convention. The design is paired: the same position appears at every level of a factor. Positions are disjoint from 67 inventoried sources, including the named public datasets and a 6.1-million-puzzle snapshot. A 100-group partition is unscored by us, and its image hashes are committed before the images are released.

2. Three documented evaluation failures (Section 4), each stated at the strength its evidence supports. A leaderboard-selected pipeline scores below its own member on 16 of 18 splits and on three further cohorts (a descriptive comparison). Two named holdouts overlapped the training mixes (a provenance failure). One renderer expression, the board colour convention, accounts for most of the per-square collapse of a state-of-the-art recogniser that had been read as a domain shift (a paired intervention).

3. A matched-position, fixed-update adaptation study (Section 5) with a preregistered real-photo retention gate. It reports what a small budget of renders changes: most of the zero-shot gap, and most of the convention gap when both conventions are rendered. It also reports what it does not resolve: the lowest elevation band reaches only 14.5 exact after the largest budget, and 11–23 real-photo boards are lost at every budget.

4. A prepared release with receipts: generator, manifests, scoring and diagnostic tools, figure pipeline, licence register, and the test-split hash list (Section 8).

What we do not claim. No new architecture and no state of the art. Results on the development partition are results on our own render lineage; on their own they say nothing about real-world robustness. The public real-photo sets we report were scored during development, and we say so where each number appears. We have not isolated the cause of the leaderboard-selected pipeline's failure; that needs an intervention we did not run.

Writing assistance. Parts of the manuscript were drafted with Codex (OpenAI) and Claude (Anthropic) from the project's records and reports; the author verified every number against its receipt and is responsible for the text.

2 Related work

Chess-position recognition. Classical pipelines detect board lines, rectify the board, extract squares and classify pieces Wölflein and Arandjelović [2021]. ChessReD Masouris and van Gemert [2024] showed that direct end-to-end prediction from smartphone photographs is possible and that exact-board accuracy is far harder than per-square accuracy. ChessQueries Seytre [2026a] decodes 64 learned square queries against a DINOv2 encoder Oquab et al. [2023] and reports 99.5 exact on ChessReD, 98.5 on ChessCog, 87.1 on SLCC and 87.6 on CVChess; we reproduce all four within 0.4 points through the authors' code path and use the release checkpoint as our external baseline throughout. That paper also reports few-shot low-rank adaptation (LoRA) and query-attention analysis; our contribution is not adapter tuning but a matched-position, fixed-update protocol with a preregistered real-photo retention gate, and a convention factor its analysis does not report varying. For Fenify Spears [2026], the report describes direct board prediction with a mobile encoder and whole-game decoding from video with a legal-move beam search. ChessRender360 Kojić [2024] is the closest synthetic dataset; its public record describes randomised cameras, materials, backgrounds and lighting, and does not document a paired single-factor protocol, a convention factor, or an audit of position overlap with other public pools.

Sim-to-real evaluation. Domain randomisation Tobin et al. [2017] treats rendering as a way to cover factor combinations that are rare or expensive to photograph. Our generator is a small instance of that idea aimed at evaluation rather than training: each split varies one factor, and the paired design lets a factor effect be read as a within-position difference.

Benchmarks that mislead. Adaptive reuse of a held-out set can inflate reported accuracy Blum and Hardt [2015], Dwork et al. [2015], and re-collected test sets can show drops that are attributable to harder sampling rather than to adaptivity Recht et al. [2019]; both mechanisms are relevant here and we do not separate them. Training and evaluation contamination has been studied directly for language models Oren et al. [2024]. Our contribution is not a new account of either but three concrete instances in one perception pipeline, one of them with a paired counterfactual.

3 Controlled generator and paired benchmark

3.1 Renderer contract

Each scene is rendered at 1920×1080 pixels from a procedural board, a piece family with a recorded asset grant, a perspective camera, and explicit lighting and material parameters. The manifest stores the position in Forsyth–Edwards Notation, the camera state, the factor assignment, and the projected board corners. Labels therefore come from the scene description rather than from a detector or a recognizer. The labels describe the latent scene state; they do not certify that every occupied square is visually identifiable in a single image under occlusion.

The renderer treats board colour convention as an explicit scene variable. The reference condition makes a1 dark; its paired stress condition makes a1 light while retaining the position, camera, geometry, palette, and framing. This intervention changes the physical checkerboard colours, not the label order or the image orientation.

3.2 Paired factor design

The benchmark varies four elevation bands, eight piece geometries, six non-reference material schemes, two framing ranges, and two board conventions. These settings form factor panels rather than a full Cartesian product: each non-reference level is paired with the appropriate reference scene, and every scene has both convention mates. A palette treatment is a supplied material scheme: it changes board and piece colours and may also change roughness and metalness. It is not an RGB-only intervention. The reference uses the highest elevation band, Poly Haven geometry, standard framing, and the separate reference material scheme. Table 1 gives the registered levels; Figure 1 summarizes the data path and the pairing structure. These are contrasts conditional on the reference settings; the design does not estimate all factor interactions.

Table 1: Paired factor panels, not a full factorial cross-product.

Factor	Levels	Values
Elevation	4	08_15, 15_25, 25_45, 45_60
Geometry	8	deplorable, killgorack, lyricsz, nailfighter, omie, phuong, pjhanzlik, polyhaven
Palette	6	aubergine_periwinkle, cobalt_porcelain, indigo_lavender, navy_coolrose, plum_mintgreen, wine_iceblue
Framing	2	compact, standard
Convention	2	a1 dark, a1 light

The design contains 500 independently sampled position groups. A fixed split assigns 400 groups to development and 100 to the sealed final test, with every view of a group kept in the same partition. Applying the 20 factor cells at both conventions yields 20,000 cell-images: 16,000 development images and 4,000 sealed images. Three reference cells deliberately reuse identical scene configurations, so the benchmark contains 18,000 unique PNG hashes. Neither cell-images nor repeated render hashes are independent board samples; the independent sampling unit remains the position group.

Each group contributes one position from a separately simulated legal random-play game. The generator deliberately mixes dense and sparse positions and biases move selection toward captures. It is not a sample of human game frequencies. Exact and symmetry checks do not exclude near-duplicate positions or shared opening structure.

3.3 Position isolation and final-test firewall

Candidate positions were rejected on exact placement and under the eight dihedral board symmetries against 67 inventoried sources. The exclusion basis contains 6,199,823 distinct placements and includes the 6,100,960-row Lichess puzzle snapshot, prior render cohorts, training pools, and the named public datasets. The benchmark and the separate adaptation pool are mutually exact- and symmetry-disjoint. The audit treats material-only similarity as a descriptive statistic, not as positional leakage. These checks concern canonical piece placement against the supplied inventory. They do not establish coverage of every public dataset, unknown foundation-model pretraining, or the complete historical training lineage.

The final-test partition has 100 position groups and 4,000 cell-images. Its hashes and integrity metadata are frozen, but no recognition, localization, attention, timing, or training result from that partition is reported here. All model results in this paper use the development partition unless

stated otherwise. Here "sealed" names our no-scoring rule, not an access-control or cryptographic barrier: releasing complete manifests, seeds, assets and a deterministic generator would permit reconstruction of those images.

3.4 Corner ground truth

The projected corners are produced by the same camera projection that renders the scene. This avoids circular detector-derived ground truth. Validation compares the analytic projection with renderer-recorded slab bounds and uses asymmetric fixtures to test named-corner identity rather than only cyclic self-consistency. The resulting corners support both perspective rectification and detector-error measurements.

3.5 Board conventions

The convention audit examines one representative image for each of the 56 image sources and records named light corners together with loader behavior. The project's earlier procedural lineage is consistently a1 light in the sampled records, while the independently anchored public representatives for ChessReD, ChessCog, CVChess, and ChessRender360 are a1 dark. The sampled SLCC frame has no independent named-corner anchor, so its physical convention remains unresolved. This is a source-sample audit, not a whole-dataset guarantee. None of the reviewed loaders mirrors the position labels to compensate for board colour. Appendix C gives the complete table.

3.6 Metrics and aggregation

The primary benchmark results report exact-board and per-square class accuracy. Exact-board accuracy requires every square label to match the target placement. The separate release-checkpoint diagnostic also archives occupied-square class accuracy, binary occupancy accuracy, confusion counts, and wrong-square histograms; these are not measurements of the adapted checkpoints. Pooled benchmark scores give equal weight to the registered cells, including the repeated reference cells, rather than to distinct image hashes. Bootstrap comparisons resample position groups with all their views together; the fixed synthetic population and protocol still limit generalization.

For the earlier, unpaired benchmark, corner error is the mean Euclidean vertex displacement after choosing the best cyclic alignment, divided by the mean projected ground-truth edge length. Missing detections and vertices extending more than two frame widths or heights outside the canvas are excluded from its conditional summaries and counted separately. This normalization removes global image scale, not projective foreshortening, and does not test named-corner orientation. The earlier elevation plot is descriptive across different positions, not a paired factor-effect estimate from the new benchmark.

4 Three ways the evaluation misled us

Each subsection states what we believed, what the record shows, and the correct statement. The three cases differ in kind. The first is a descriptive comparison between systems that differ in more than one way. The second is a provenance failure. Only the third has a paired intervention that isolates a cause. Numbers for the first two come from the earlier benchmark generation (v1: 18 splits, 5,800 frames, same generator lineage, a1-light convention throughout) and from public real-photo sets. They are hash-pinned historical summaries, not original run receipts; Appendix F records the gap. Numbers for the third come from the paired parity probes and from benchmark v2.

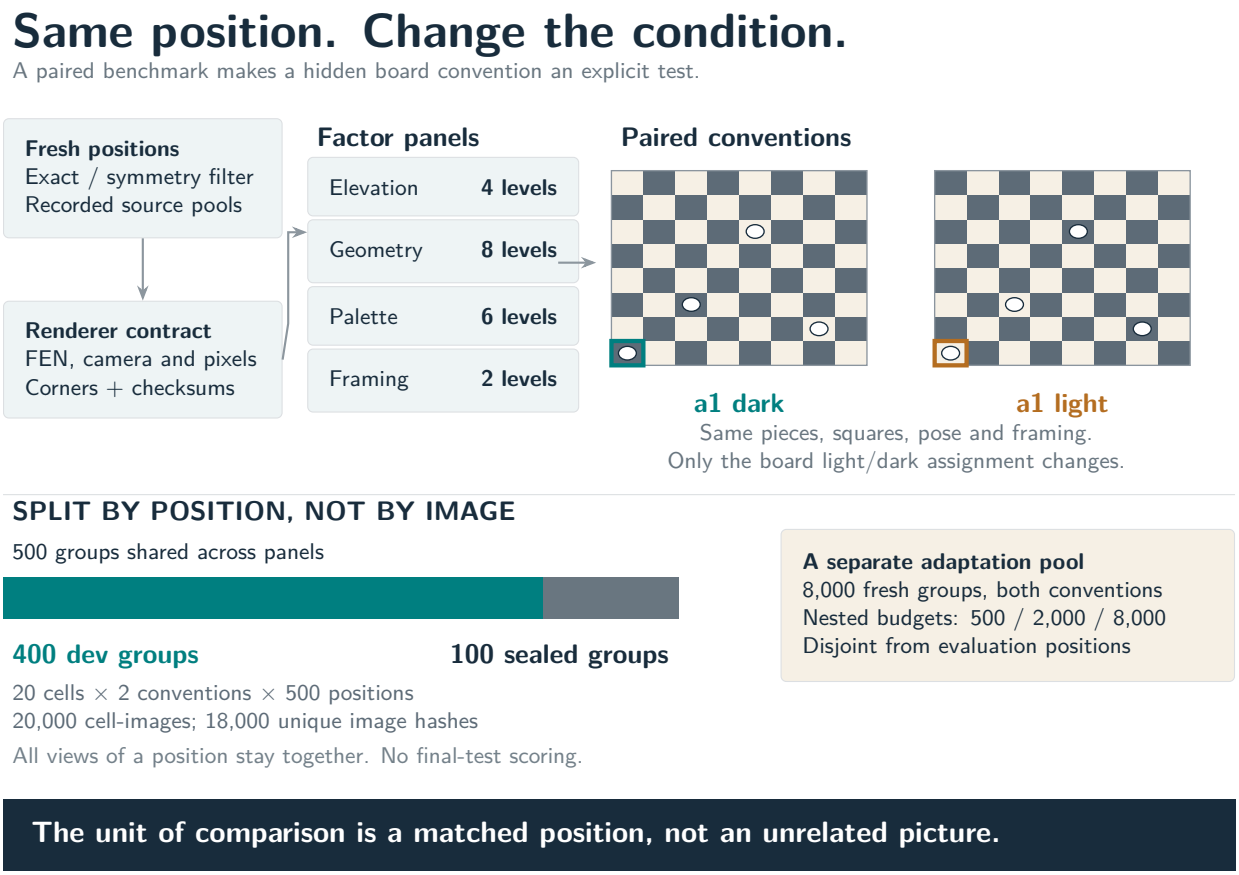


Figure 1: Paired benchmark design. The boards are vector illustrations, not experimental frames. Each panel varies one factor; the convention is paired within every panel.

The probe, baseline and holdout-check result summaries are in the evidence index (Appendix F); per-image prediction files are not.

4.1 A leaderboard-selected pipeline scores below its own member

The system we had shipped routed each image, by the detected board's apparent angle, to one of three members: a full-frame recogniser initialised from ChessQueries (the *raw member*) and two recognisers that operate on a rectified board crop (the *warp-high* and *warp-low* members). The router and its thresholds had been chosen on a private leaderboard. On our 18 controlled splits the shipped ensemble scores below its own raw member on 16, by up to 32.3 exact-board points on one piece family. The raw member alone reaches 100.0 on twelve splits (Figure 2). Section 6 reports the latency of both systems.

Three further comparisons point the same way. Each carries a caveat that makes it a development number, not a replication. SLCC test: 52.45 against 74.73, on 368 boards from 13 videos, and the raw member had SLCC training data in its mix. CVChess: 1.42 against 98.86, on 352 photographs of only 88 distinct positions. A held-out block of the training scenes: 99.30 against 100.00, on held-out move blocks of the same games and scenes as training. On SLCC, above the routing threshold the shipped ensemble drops the raw member and uses only the two warp members. Run alone on all 368 boards, those members produce eight or more wrong squares on 234 and 236 boards. That is consistent with failures in the detector and rectification stages on broadcast crops. We ran no oracle-corner ablation on that cohort, so localisation and recognition failures are not separated.

Routing, member composition and training lineage all differ between the shipped ensemble and the raw member, so this is not an isolated routing effect. We also have no intervention on leaderboard exposure, so the reversal cannot be attributed to adaptive reuse rather than to distribution shift between the leaderboard and these cohorts. What the record supports is this: a pipeline selected on one private cohort scored below its own simpler member on the pooled v1 benchmark (91.8 against 98.5 exact over 5,800 frames) and on each of the three further cohorts. The two exceptions among the 18 splits are the two lowest elevation bands, where the pipeline scores 88.8 against 88.6 and 98.0 against 97.0.

4.2 Two holdouts contaminated by construction

An identifier-level leak check over the training mixes and every named holdout found that two holdouts derived from ChessRender360 (1,635 and 1,263 images) shared image identifiers with the mixes trained on them: 1,446 and 1,128 identifiers, 2,398 distinct, 2,574 counting both sets (5,148 mix-by-identifier events over two mixes). Both holdouts were subsets of the same ten thousand rectified images. The check reports `clean: false`. The documented model-selection replay does not use those two holdouts, so the recorded selection decisions were not influenced by them. That is not a proof that no earlier or unrecorded decision was. A set can be named a holdout and still be training data. A check that runs on identifiers and positions, not on directory names, is what caught it.

4.3 A convention confound mistaken for a domain shift

The ChessQueries release checkpoint produces zero exact boards on every one of the 18 v1 splits. Its per-square accuracy of 52.7–56.7 is 4.6–9.0 points *below* a constant all-empty prediction, which also produces zero exact boards on these positions. The model places pieces on the wrong squares; it does not default to empty. The same checkpoint through the same code path scores 99.62 on

ChessReD and 98.54 on ChessCog, which is also rendered. The failure appeared specific to our rendering lineage. A crop diagnostic fed the checkpoint the first 150 boards of each of three splits (450 in all) four ways: full frame, bounding-box crop, square-padded crop, and a perspective warp to a square. None recovered an exact board under our convention, and per-square accuracy moved by at most 1.2 points from full frame. At that point our own notes read "the shift is the piece and board appearance itself."

It was not. Our renderer, following the convention of the data it had been matched to, makes a1 a *light* square. The laws of chess place a dark square at a1 Fédération Internationale des Échecs [2023]. Every public chess-recognition source we could resolve follows that convention (Table 5). The audit checks one representative image per source; the SLCC sample has no printed coordinates, so its corner is recorded as unresolved. A preregistered paired probe changed exactly that one renderer expression and nothing else, with byte-identical controls. On 192 positions in two disjoint waves the checkpoint went from 0/192 to 18/192 exact (+9.38 points, 95% paired bootstrap $[5.73, 13.54]$) and from 54.36 to 85.24 per-square (+30.88, $[28.91, 32.76]$), removing 67.65% of wrong squares. Replicated on 288 positions across six other piece families, the effect is 0/288 to 50/288 exact and 55.31 to 87.49 per-square. Swapping the model's learned two-class query-colour indices Seytre [2026a] in memory recovers 20.7 per-square points on a1-light images but no exact board, and costs 2.9 points on a1-dark images. This is consistent with the colour prior contributing to the failure; it does not isolate the mechanism. On the development partition of benchmark v2 the release checkpoint scores 15.48 exact pooled under the a1-dark convention, ranging from 0 to 34.25 across the twenty factor cells, and 0.0 in every cell under a1-light (Figure 3, left). One real-board check points the same way. The same ten positions were photographed on one physical board with a dark a1, and on a later day, with a different phone and photographer, with a light a1. The release checkpoint scores 76.25 and 57.97 per-square on the two sets of ten, with no exact board in either, and the light arm scores lower on 8 of 10 positions. The arms differ in more than parity, so this is a descriptive check, not a paired probe.

The corrected statement is about per-square accuracy. The convention accounts for about two thirds of the wrong squares, and most boards still fail exactly after the correction (18/192 and 50/288). On this cohort the residual is family-associated: two families stay at 0/48 exact while another reaches 19/48. Positions differ between families here, so this describes the unchanged release checkpoint; it is not a paired geometry effect, and it says nothing about what adaptation later fixes. The paired geometry cells of v2 and the adaptation runs are the place to read those (Section 5). The trap had been written in our own research notes before the benchmark was built, and we missed it anyway. Making the convention an explicit factor let it be measured instead of inferred, which is why v2 carries it as its fifth axis.

A bounded follow-up on the same 192 positions found that, under the standard convention, a ground-truth square crop raised exact boards from 18 to 27 while lowering per-square accuracy from 85.24 to 84.24. Framing is therefore not irrelevant once the convention is corrected. The interaction is a trade-off measured on development positions with privileged localisation, not a deployable improvement.

5 Matched-position adaptation and retention cost

5.1 Protocol

We adapt only the encoder with the upstream low-rank modules [Hu et al., 2022, Seytre, 2026a]. The registered configuration uses rank 8, alpha 16, dropout 0.05, AdamW at a learning rate of 10^{-4} with zero weight decay, no scheduler, and no warm-up. The base encoder and the complete readout

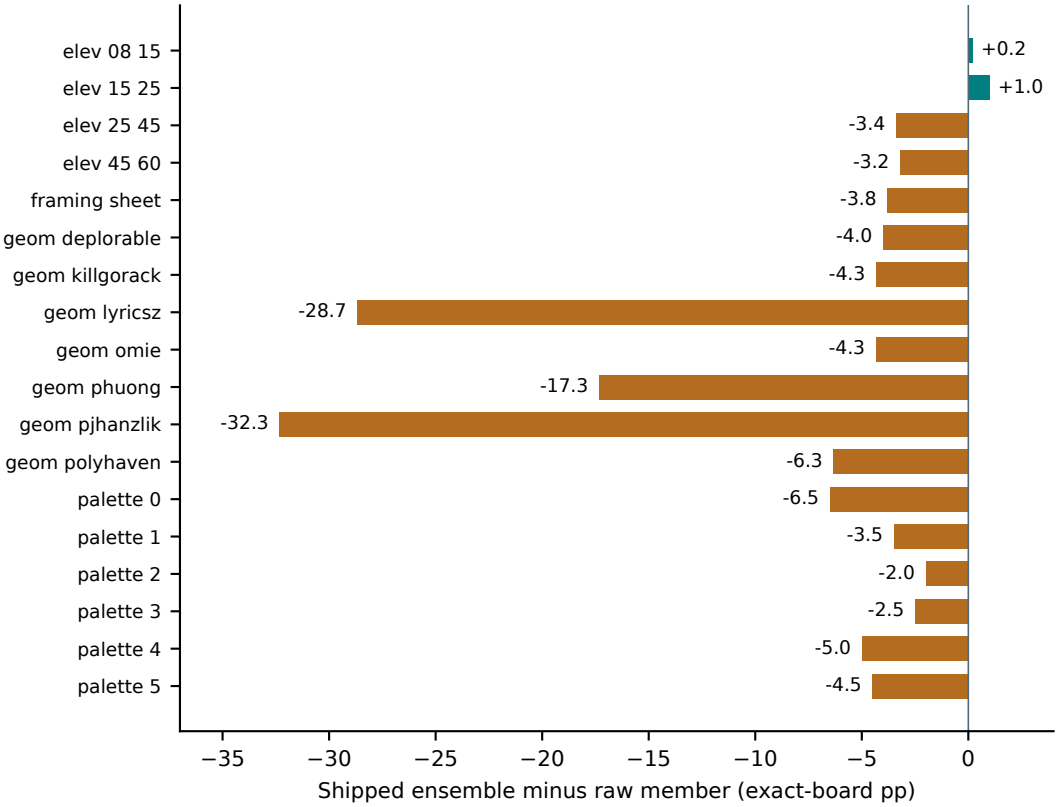


Figure 2: Descriptive pipeline comparison on the earlier controlled splits. Member composition and routing both change; the difference is not an isolated routing effect. Historical aggregates are hash-pinned; original run-completion receipts are unavailable.

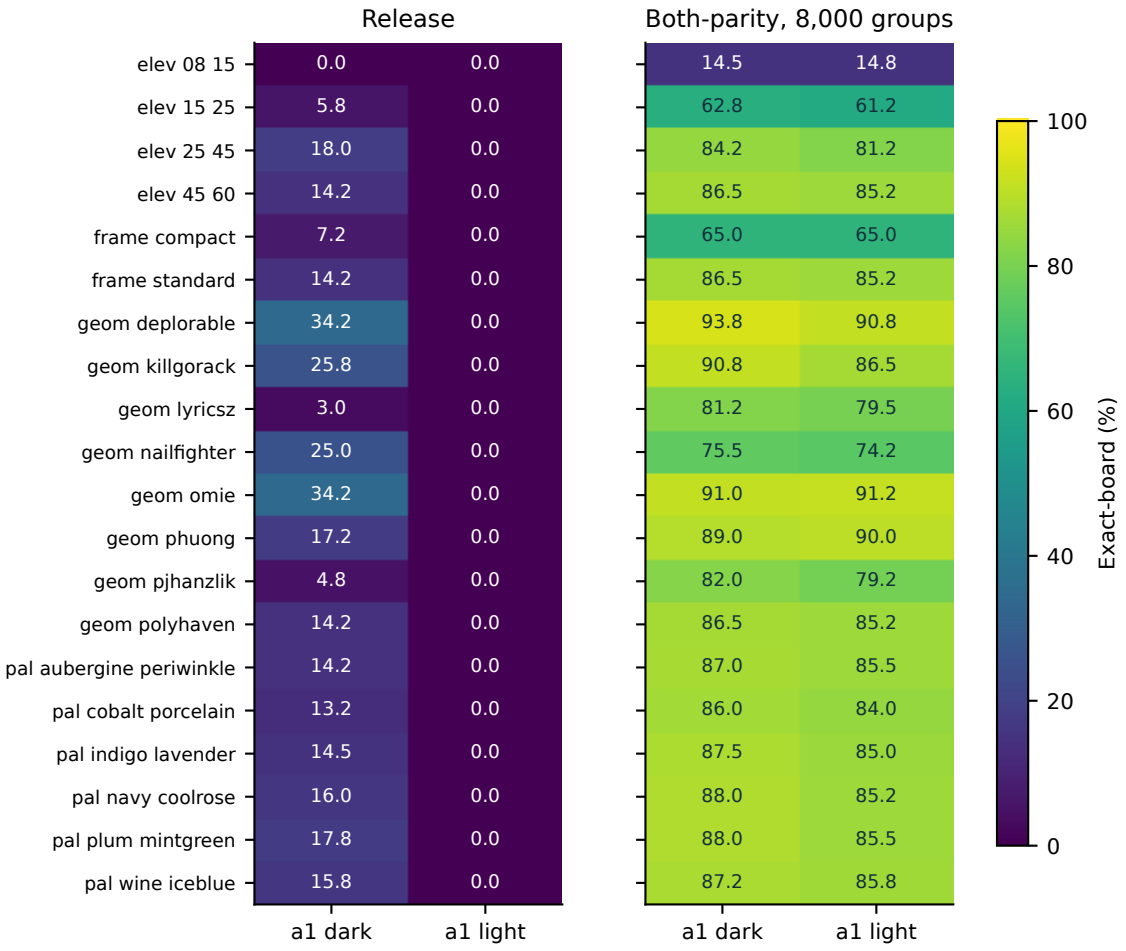


Figure 3: Exact-board accuracy by development cell and convention. Left: released weights. Right: the largest both-convention adaptation run.

remain frozen. Every run uses batch size 6, bf16 autocast, and exactly 1,500 optimizer updates, with no image augmentation, early stopping, or development-selected checkpoint.

Training uses a separate pool of 8,000 fresh position groups, each rendered at both board conventions. The 500-, 2,000-, and 8,000-group budgets are nested prefixes of one fixed permutation. The dark-only and both-convention arms use the same positions and the same number of optimizer updates, but not the same number of distinct images: a both-convention pool contains twice as many available views. Unlike the evaluation panels, adaptation examples combine factor levels with balanced marginals. The pool includes the reference material scheme as well as the palette treatments; matching the supported levels does not make the joint training and evaluation scene distributions identical.

Each run makes 9,000 image presentations. Thus the largest both-convention pool cannot be covered in full under the fixed step cap. The budget denotes available position groups, not measured unique examples seen or complete epochs. The experiments compare matched positions and fixed optimizer exposure rather than equal distinct-image counts.

All checkpoints are evaluated on the same 16,000 development cell-images from 400 position groups. The separate photo retention set contains 2,129 boards; the preregistered pass threshold is 99.12% exact-board accuracy. The sealed final test is not evaluated. The middle budget uses three fixed training seeds, while the other budgets use one, and every registered result is retained.

5.2 Adaptation curve

The release checkpoint begins at 15.48% exact accuracy on a1-dark development cells and 0.00% on a1-light cells. Dark-only adaptation raises a1-dark accuracy to 64.70% with 500 groups and 79.70% with 8,000 groups, but a1-light exact accuracy remains 0.00% at every dark-only budget. At 8,000 groups, the both-convention arm reaches 80.65% on a1-dark cells and 79.03% on a1-light cells. Figure 4 shows all seeds and budgets; the full numeric matrix appears in Table 2.

Table 2: All fixed-step runs. Synthetic cells are paired; photo counts use the same retention set.

Run	Dark (%)	Light (%)	Photo (%)	Photo exact	Gate
release	15.48	0.00	99.62	2121	reference
b500_dark_s303	64.70	0.00	99.11	2110	fail
b500_both_s303	66.80	64.20	99.01	2108	fail
b2000_dark_s303	77.85	0.00	99.01	2108	fail
b2000_both_s303	74.24	71.99	98.54	2098	fail
b2000_dark_s304	76.83	0.00	99.06	2109	fail
b2000_both_s304	73.64	72.24	98.92	2106	fail
b2000_dark_s305	77.80	0.00	98.97	2107	fail
b2000_both_s305	76.63	75.90	98.59	2099	fail
b8000_dark_s303	79.70	0.00	99.11	2110	fail
b8000_both_s303	80.65	79.03	98.59	2099	fail

The standard-convention difference between arms depends on budget. Both-convention minus dark-only accuracy is +2.10 points at 500 groups, −2.66 points for the three-seed mean at 2,000 groups, and +0.95 points at 8,000 groups. There is no budget-independent standard-convention penalty. These are point estimates, not paired between-arm confidence intervals.

As one measure of position-sampling uncertainty, the largest both-convention run improves the cell-weighted score pooled over both conventions by 72.10 points relative to release, with a stored

95% group-bootstrap interval of [69.90, 74.21]. This is not an interval for the arm contrast or for unseen real photographs. The machine-readable report retains each run's pooled and per-cell intervals and gained/lost counts; the tables print point estimates.

No adapted checkpoint passes the registered photo-retention gate. Across the ten runs, exact accuracy on that set falls from the 99.62% release reference to between 98.54% and 99.11%. This corresponds to 11–23 additional incorrect boards. These are net changes in exact-board counts on the same photo set, not counts of all previously correct boards that became incorrect. The loss is not monotone in synthetic data budget; these measurements do not identify its cause.

Recovery is not uniform across the measured cells. Under the largest both-convention run, the lowest-elevation a1-dark cell reaches only 14.50% exact accuracy while the reference-elevation cell reaches 86.50%. Geometry effects also remain visible. Figure 3 reports all factor cells for the release checkpoint and the largest both-convention run.

5.3 Reading the intervention

The experiments support three bounded conclusions. First, much of the release checkpoint's synthetic gap is reducible with matched rendered data. Second, the one-sided runs produce no exact boards under the other convention in this protocol; that does not imply zero per-square learning. Third, adding the convention mate is sufficient to remove most of that asymmetry, but it does not satisfy the real-photo retention criterion and does not resolve low-elevation error. These are development results for one architecture and one fixed adapter recipe, not evidence of general robustness.

5.4 Residual error characterization

A separate preregistered diagnostic examines 288 canonical a1-dark boards from six geometry families using the unchanged release checkpoint, not a LoRA-adapted model. The families use different positions, so their comparison does not isolate geometry. Lyricsz and PJ Hanzlik each have 0 exact boards among 48. For these families, binary presence errors account for 66.86% and 73.71% of wrong squares, respectively; the remainder are wrong occupied classes. This decomposition describes the failure but does not establish its cause.

The accompanying attention sample was chosen without reference to prediction outcomes and retains all 64 final-decoder query maps for 30 boards. The maps use head averages and per-query display normalization. Appendix E shows fixed examples. The capture used the preregistered precision amendment after an initial forward-equivalence failure; its numerical checks were passed before interpretation. Attention is treated only as descriptive localization evidence; no attention intervention or mechanism test was performed.

6 Inference cost

We measure synchronized native-inference latency on an NVIDIA RTX 3090, with no competing GPU compute process detected, at a fixed 270 W power limit. A fixed development subset supplies 500 images from 250 factor-cell scenes. Each system runs in a fresh process with batch size 1 and native FP16 autocast. The harness excludes file decode and 30 warm-up images, while retaining resize, normalization, device transfer, model execution, routing, warping, test-time augmentation, host transfer, and notation construction whenever the native pipeline uses them.

The release checkpoint has median latency 64.41 ms per image. The raw member requires 79.69 ms, while the shipped ensemble requires 195.52 ms. The ten adapted checkpoints lie between

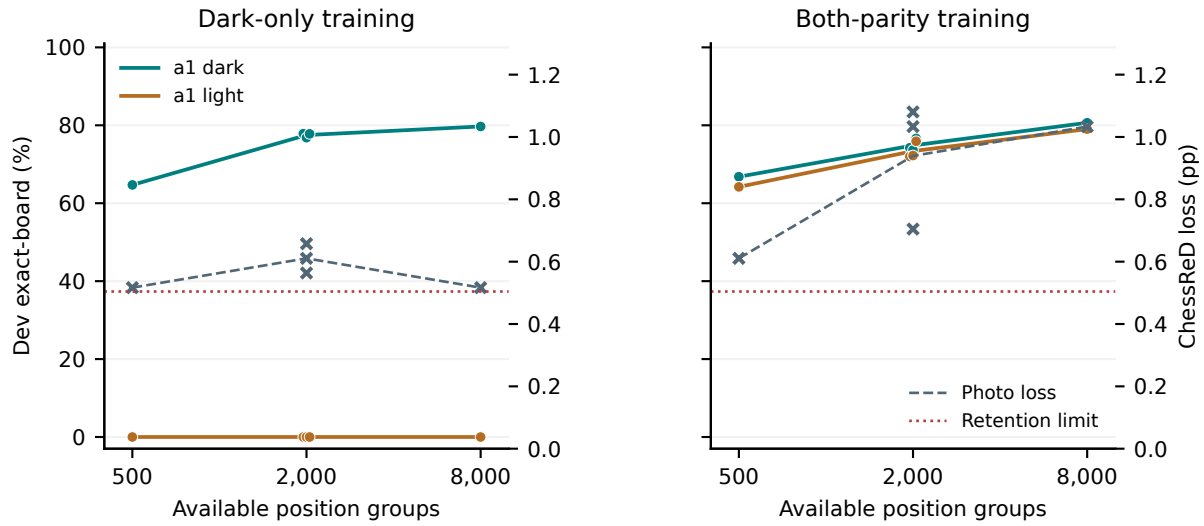


Figure 4: Fixed-step adaptation. Points show all seeds; lines join budget-wise means. The right axis measures loss on the same real-photo retention set.

71.50 and 73.12 ms, so their measured median overhead relative to the release checkpoint is 7.09–8.71 ms in this harness. Table 3 reports medians and ninetieth percentiles for every system, and Figure 5 visualizes the same receipt-bound values.

These measurements compare complete native inference paths rather than equal floating-point workloads. In particular, the ensemble and its members differ in routing, member composition, warping, and test-time augmentation. The timing experiment supports no accuracy claim and should not be extrapolated beyond the recorded hardware, software, and input pipeline. Each model has one timed process; percentiles describe variation across input images, not run-to-run repeatability. No repeated-process confidence interval or whole-project rendering/training cost is estimated by this table.

7 Limitations and threats to validity

Development data on our own lineage. Every v2 model result in this paper is on the 400-group development partition of a benchmark rendered by our own generator. Adaptation trains on a joint mix of factor levels while evaluation reads conditional panels, so a cell result is not the effect of that factor alone on a model trained without it. The 100-group partition is unscored by us and its hashes are committed before release; after release, the manifests, seeds, assets and generator make its pixels reconstructible, so it is an author-unscored split, not a hidden challenge set.

Real photographs were scored during development. SLCC, CVChess and ChessReD were each scored several times while the systems were being built, and ChessCog is itself rendered. Their numbers are development numbers. No untouched real-photo anchor is part of this version.

Historical provenance. The v1 spine (Section 4.1) rests on hash-pinned summary reports whose original execution receipts were not retained; the v1 freeze also records a stale source revision.

Table 3: Synchronized single-image latency on the same isolated device.

System	Images	Median (ms)	P90 (ms)
ChessQueries release	500	64.41	65.72
shipped ensemble	500	195.52	244.22
raw member	500	79.69	81.07
warp-high member	500	195.23	197.55
warp-low member	500	168.98	170.85
LoRA dark n500 s303	500	71.88	72.64
LoRA both n500 s303	500	72.40	73.32
LoRA dark n2000 s303	500	72.29	73.08
LoRA both n2000 s303	500	71.50	72.50
LoRA dark n2000 s304	500	71.86	72.95
LoRA both n2000 s304	500	72.80	73.68
LoRA dark n2000 s305	500	72.50	73.48
LoRA both n2000 s305	500	73.12	74.03
LoRA dark n8000 s303	500	72.40	73.26
LoRA both n8000 s303	500	72.66	73.57

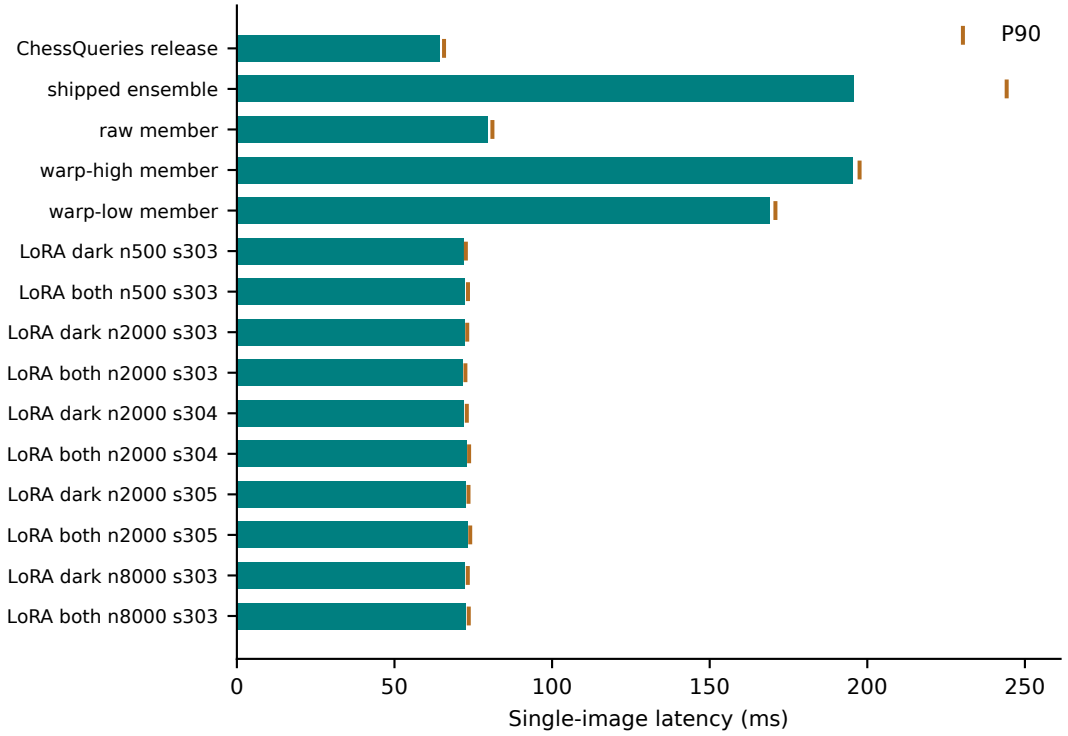


Figure 5: Single-image synchronized latency. Bars show medians; markers show upper-decile latency. This experiment does not compare accuracy.

Appendix F lists the receipts and result summaries in this package; the per-image prediction files behind Section 4 are not among them. We did not rebuild those runs.

Independent units. Benchmark v2 has 400 development position groups, not 16,000 independent boards; cells share positions by design, and reference views are shared across panels. Every interval is a paired bootstrap over position groups. Seed spread at the 2,000-position budget is about one exact point in the dark-only arm and about three in the both-convention arm.

One architecture, one adapter, matched positions not matched images. The adaptation study uses one recogniser and the authors' LoRA implementation with upstream optimiser settings and a fixed update count. Arms match position pools and optimiser presentations, not the number of available rendered images. The retention-gate failure does not show that every adaptation method must fail, and the cause of the retention loss is unresolved.

Conditional statistics. Detector localisation summaries exclude divergent detections, and that fraction is itself factor-dependent; we publish it beside every summary and make no "no effect" claim without a predefined margin.

Licensing and packaging. Seven of eight piece families carry verified CC0 source grants. One is CC-BY with its version not displayed by the source page; it is credited by name and source, and we apply the CC BY 4.0 attribution terms to it. Two families with no licence grant were excluded from every split. Project code, vendored third-party code, rendered data, the upstream base weights (non-commercial) and our adapters are separate licensing questions. The first three now have licence files in the prepared package; the base weights will be linked rather than redistributed, and the adapters are withheld.

8 Release package

The release package is a sanitised export of the author's repository. Identifiers of private machines and paths are replaced, and digests of the unredacted form of rewritten files are withheld. The package contains the generator and renderers, the v2 manifests and freeze receipts, the adaptation-pool manifests, the scoring and diagnostic tools, the figure pipeline that produced every generated figure and table in this paper from receipts, the licence register (Tables 6 and 7), and the convention table. The paper source and PDF are in the package. The licence map splits the package by material class: the generator and renderers are source-available under a noncommercial licence (PolyForm Noncommercial 1.0.0); the scoring, diagnostic and export tools are Apache-2.0; manifests, receipts, hash lists and rendered images are CC BY-NC 4.0, with attribution to each piece family's creator; the paper text, figures and tables are CC BY-NC 4.0 and the paper's build scripts are Apache-2.0; an added permission lets anyone, companies included, evaluate models on the benchmark and publish the scores; vendored third-party code keeps its upstream notices; the upstream base weights are non-commercial and are linked, not redistributed; the adapters derive from those weights and are withheld.

The manifest rows of the 100-group test partition (positions, seeds, cameras and image hashes) are in the package. Its images are not, and the author scored no model on it. Anyone can rebuild its pixels from the manifest and the generator, so it is unscored, not hidden (Section 7). Images published later can be checked against the image hashes in the manifest.

Every number in the generated tables carries a source comment naming its receipt. The package README gives a bounded reproduction path: install, re-render four benchmark images and compare their hashes, then score them with the public baseline on CPU. It does not re-render the full benchmark and does not rerun adaptation training. The earlier private systems are reported from their pinned summaries and are not redistributed. Private material used during development, including the leaderboard those systems were tuned on, is not part of the release.

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A Full cell results

Table 4 contains every development factor cell for the release checkpoint and every registered adaptation run. Entries are generated directly from the copied report receipt; no unsuccessful run or cell is removed. Percentages in these tables use decimal half-up rounding to two places; retention decisions use the unrounded recorded values. These are point-estimate tables. The accompanying machine-readable report also contains the registered paired intervals and gained/lost counts; no between-arm interval is claimed by the plotted seed spread.

Table 4: Complete development cell results for the release and each adaptation run.

Run	Cell / convention	N	Exact (%)	Square (%)
release	elevation:08_15/dark	400	0.00	75.04
release	elevation:08_15/light	400	0.00	57.89
release	elevation:15_25/dark	400	5.75	88.21
release	elevation:15_25/light	400	0.00	56.19
release	elevation:25_45/dark	400	18.00	87.28
release	elevation:25_45/light	400	0.00	53.94
release	elevation:45_60/dark	400	14.25	84.19
release	elevation:45_60/light	400	0.00	52.89
release	framing:compact/dark	400	7.25	85.01
release	framing:compact/light	400	0.00	52.18
release	framing:standard/dark	400	14.25	84.19
release	framing:standard/light	400	0.00	52.89
release	geometry:deplorable/dark	400	34.25	86.95
release	geometry:deplorable/light	400	0.00	53.63
release	geometry:killgorack/dark	400	25.75	85.27
release	geometry:killgorack/light	400	0.00	53.81
release	geometry:lyricsz/dark	400	3.00	83.43
release	geometry:lyricsz/light	400	0.00	55.74
release	geometry:nailfighter/dark	400	25.00	87.52
release	geometry:nailfighter/light	400	0.00	54.04
release	geometry:omie/dark	400	34.25	88.35
release	geometry:omie/light	400	0.00	54.60
release	geometry:phuong/dark	400	17.25	86.04
release	geometry:phuong/light	400	0.00	54.87
release	geometry:pjhanzlik/dark	400	4.75	80.97
release	geometry:pjhanzlik/light	400	0.00	54.66
release	geometry:polyhaven/dark	400	14.25	84.19
release	geometry:polyhaven/light	400	0.00	52.89
release	palette:aubergine_periwinkle/dark	400	14.25	84.75
release	palette:aubergine_periwinkle/light	400	0.00	52.86
release	palette:cobalt_porcelain/dark	400	13.25	84.39
release	palette:cobalt_porcelain/light	400	0.00	53.40
release	palette:indigo_lavender/dark	400	14.50	84.73
release	palette:indigo_lavender/light	400	0.00	52.99
release	palette:navy_coolrose/dark	400	16.00	84.61
release	palette:navy_coolrose/light	400	0.00	53.44
release	palette:plum_mintgreen/dark	400	17.75	85.08
release	palette:plum_mintgreen/light	400	0.00	52.19
release	palette:wine_iceblue/dark	400	15.75	84.54

Run	Cell / convention	N	Exact (%)	Square (%)
release	palette:wine_iceblue/light	400	0.00	52.81
b500_dark_s303	elevation:08_15/dark	400	7.00	93.39
b500_dark_s303	elevation:08_15/light	400	0.00	66.27
b500_dark_s303	elevation:15_25/dark	400	49.00	98.39
b500_dark_s303	elevation:15_25/light	400	0.00	60.79
b500_dark_s303	elevation:25_45/dark	400	70.50	99.03
b500_dark_s303	elevation:25_45/light	400	0.00	56.84
b500_dark_s303	elevation:45_60/dark	400	68.00	98.98
b500_dark_s303	elevation:45_60/light	400	0.00	55.80
b500_dark_s303	framing:compact/dark	400	51.50	98.40
b500_dark_s303	framing:compact/light	400	0.00	56.76
b500_dark_s303	framing:standard/dark	400	68.00	98.98
b500_dark_s303	framing:standard/light	400	0.00	55.80
b500_dark_s303	geometry:deplorable/dark	400	84.75	99.43
b500_dark_s303	geometry:deplorable/light	400	0.00	55.73
b500_dark_s303	geometry:killgorack/dark	400	81.50	99.39
b500_dark_s303	geometry:killgorack/light	400	0.00	55.97
b500_dark_s303	geometry:lyricsz/dark	400	43.50	98.40
b500_dark_s303	geometry:lyricsz/light	400	0.00	57.66
b500_dark_s303	geometry:nailfighter/dark	400	59.75	98.88
b500_dark_s303	geometry:nailfighter/light	400	0.00	56.40
b500_dark_s303	geometry:omie/dark	400	88.00	99.52
b500_dark_s303	geometry:omie/light	400	0.00	56.77
b500_dark_s303	geometry:phuong/dark	400	75.50	99.25
b500_dark_s303	geometry:phuong/light	400	0.00	56.92
b500_dark_s303	geometry:pjhanzlik/dark	400	55.25	98.68
b500_dark_s303	geometry:pjhanzlik/light	400	0.00	57.50
b500_dark_s303	geometry:polyhaven/dark	400	68.00	98.98
b500_dark_s303	geometry:polyhaven/light	400	0.00	55.80
b500_dark_s303	palette:aubergine_periwinkle/dark	400	69.50	99.03
b500_dark_s303	palette:aubergine_periwinkle/light	400	0.00	55.46
b500_dark_s303	palette:cobalt_porcelain/dark	400	70.50	98.94
b500_dark_s303	palette:cobalt_porcelain/light	400	0.00	55.84
b500_dark_s303	palette:indigo_lavender/dark	400	71.50	99.02
b500_dark_s303	palette:indigo_lavender/light	400	0.00	55.56
b500_dark_s303	palette:navy_coolrose/dark	400	69.25	98.97
b500_dark_s303	palette:navy_coolrose/light	400	0.00	55.67
b500_dark_s303	palette:plum_mintgreen/dark	400	73.50	99.14
b500_dark_s303	palette:plum_mintgreen/light	400	0.00	55.35
b500_dark_s303	palette:wine_iceblue/dark	400	69.50	99.00
b500_dark_s303	palette:wine_iceblue/light	400	0.00	55.32
b500_both_s303	elevation:08_15/dark	400	7.00	93.44
b500_both_s303	elevation:08_15/light	400	7.75	93.32
b500_both_s303	elevation:15_25/dark	400	47.50	98.34
b500_both_s303	elevation:15_25/light	400	43.50	98.19
b500_both_s303	elevation:25_45/dark	400	73.75	99.02
b500_both_s303	elevation:25_45/light	400	70.25	98.92
b500_both_s303	elevation:45_60/dark	400	71.50	98.84
b500_both_s303	elevation:45_60/light	400	69.75	98.80
b500_both_s303	framing:compact/dark	400	54.75	98.40
b500_both_s303	framing:compact/light	400	51.00	98.07
b500_both_s303	framing:standard/dark	400	71.50	98.84

Run	Cell / convention	N	Exact (%)	Square (%)
b500_both_s303	framing:standard/light	400	69.75	98.80
b500_both_s303	geometry:deplorable/dark	400	86.00	99.21
b500_both_s303	geometry:deplorable/light	400	82.75	99.16
b500_both_s303	geometry:killgorack/dark	400	81.50	99.14
b500_both_s303	geometry:killgorack/light	400	77.75	99.02
b500_both_s303	geometry:lyricsz/dark	400	48.25	98.00
b500_both_s303	geometry:lyricsz/light	400	47.25	97.94
b500_both_s303	geometry:nailfighter/dark	400	59.50	98.68
b500_both_s303	geometry:nailfighter/light	400	54.50	98.55
b500_both_s303	geometry:omie/dark	400	86.00	99.27
b500_both_s303	geometry:omie/light	400	83.25	99.17
b500_both_s303	geometry:phuong/dark	400	73.00	98.94
b500_both_s303	geometry:phuong/light	400	71.50	98.85
b500_both_s303	geometry:pjhanzlik/dark	400	62.25	98.63
b500_both_s303	geometry:pjhanzlik/light	400	56.50	98.47
b500_both_s303	geometry:polyhaven/dark	400	71.50	98.84
b500_both_s303	geometry:polyhaven/light	400	69.75	98.80
b500_both_s303	palette:aubergine_periwinkle/dark	400	73.75	98.93
b500_both_s303	palette:aubergine_periwinkle/light	400	72.25	98.82
b500_both_s303	palette:cobalt_porcelain/dark	400	70.50	98.85
b500_both_s303	palette:cobalt_porcelain/light	400	68.50	98.78
b500_both_s303	palette:indigo_lavender/dark	400	73.75	98.92
b500_both_s303	palette:indigo_lavender/light	400	70.75	98.85
b500_both_s303	palette:navy_coolrose/dark	400	73.25	98.91
b500_both_s303	palette:navy_coolrose/light	400	71.75	98.84
b500_both_s303	palette:plum_mintgreen/dark	400	77.25	99.01
b500_both_s303	palette:plum_mintgreen/light	400	74.00	98.88
b500_both_s303	palette:wine_iceblue/dark	400	73.50	98.91
b500_both_s303	palette:wine_iceblue/light	400	71.50	98.82
b2000_dark_s303	elevation:08_15/dark	400	15.25	95.60
b2000_dark_s303	elevation:08_15/light	400	0.00	67.24
b2000_dark_s303	elevation:15_25/dark	400	62.75	98.94
b2000_dark_s303	elevation:15_25/light	400	0.00	60.72
b2000_dark_s303	elevation:25_45/dark	400	83.25	99.43
b2000_dark_s303	elevation:25_45/light	400	0.00	56.59
b2000_dark_s303	elevation:45_60/dark	400	80.75	99.37
b2000_dark_s303	elevation:45_60/light	400	0.00	55.51
b2000_dark_s303	framing:compact/dark	400	64.50	99.00
b2000_dark_s303	framing:compact/light	400	0.00	56.48
b2000_dark_s303	framing:standard/dark	400	80.75	99.37
b2000_dark_s303	framing:standard/light	400	0.00	55.51
b2000_dark_s303	geometry:deplorable/dark	400	91.25	99.64
b2000_dark_s303	geometry:deplorable/light	400	0.00	55.05
b2000_dark_s303	geometry:killgorack/dark	400	88.75	99.54
b2000_dark_s303	geometry:killgorack/light	400	0.00	55.31
b2000_dark_s303	geometry:lyricsz/dark	400	76.25	99.26
b2000_dark_s303	geometry:lyricsz/light	400	0.00	56.97
b2000_dark_s303	geometry:nailfighter/dark	400	77.00	99.29
b2000_dark_s303	geometry:nailfighter/light	400	0.00	55.96
b2000_dark_s303	geometry:omie/dark	400	91.75	99.62
b2000_dark_s303	geometry:omie/light	400	0.00	56.17
b2000_dark_s303	geometry:phuong/dark	400	85.75	99.48

Run	Cell / convention	N	Exact (%)	Square (%)
b2000_dark_s303	geometry:phuong/light	400	0.00	56.14
b2000_dark_s303	geometry:pjhanzlik/dark	400	81.00	99.46
b2000_dark_s303	geometry:pjhanzlik/light	400	0.00	56.58
b2000_dark_s303	geometry:polyhaven/dark	400	80.75	99.37
b2000_dark_s303	geometry:polyhaven/light	400	0.00	55.51
b2000_dark_s303	palette:aubergine_periwinkle/dark	400	82.50	99.39
b2000_dark_s303	palette:aubergine_periwinkle/light	400	0.00	55.42
b2000_dark_s303	palette:cobalt_porcelain/dark	400	79.75	99.34
b2000_dark_s303	palette:cobalt_porcelain/light	400	0.00	55.85
b2000_dark_s303	palette:indigo_lavender/dark	400	83.00	99.40
b2000_dark_s303	palette:indigo_lavender/light	400	0.00	55.36
b2000_dark_s303	palette:navy_coolrose/dark	400	84.50	99.44
b2000_dark_s303	palette:navy_coolrose/light	400	0.00	55.50
b2000_dark_s303	palette:plum_mintgreen/dark	400	84.75	99.45
b2000_dark_s303	palette:plum_mintgreen/light	400	0.00	55.06
b2000_dark_s303	palette:wine_iceblue/dark	400	82.75	99.39
b2000_dark_s303	palette:wine_iceblue/light	400	0.00	55.10
b2000_both_s303	elevation:08_15/dark	400	13.75	95.38
b2000_both_s303	elevation:08_15/light	400	14.75	95.31
b2000_both_s303	elevation:15_25/dark	400	61.50	98.91
b2000_both_s303	elevation:15_25/light	400	57.00	98.75
b2000_both_s303	elevation:25_45/dark	400	79.50	99.25
b2000_both_s303	elevation:25_45/light	400	77.25	99.15
b2000_both_s303	elevation:45_60/dark	400	76.75	99.12
b2000_both_s303	elevation:45_60/light	400	73.75	99.07
b2000_both_s303	framing:compact/dark	400	61.25	98.71
b2000_both_s303	framing:compact/light	400	56.75	98.59
b2000_both_s303	framing:standard/dark	400	76.75	99.12
b2000_both_s303	framing:standard/light	400	73.75	99.07
b2000_both_s303	geometry:deplorable/dark	400	87.50	99.49
b2000_both_s303	geometry:deplorable/light	400	86.50	99.43
b2000_both_s303	geometry:killgorack/dark	400	87.00	99.46
b2000_both_s303	geometry:killgorack/light	400	84.00	99.38
b2000_both_s303	geometry:lyricsz/dark	400	77.00	99.30
b2000_both_s303	geometry:lyricsz/light	400	75.25	99.22
b2000_both_s303	geometry:nailfighter/dark	400	72.75	99.11
b2000_both_s303	geometry:nailfighter/light	400	71.75	99.11
b2000_both_s303	geometry:omie/dark	400	88.00	99.47
b2000_both_s303	geometry:omie/light	400	85.25	99.38
b2000_both_s303	geometry:phuong/dark	400	82.00	99.33
b2000_both_s303	geometry:phuong/light	400	80.25	99.29
b2000_both_s303	geometry:pjhanzlik/dark	400	79.50	99.32
b2000_both_s303	geometry:pjhanzlik/light	400	78.25	99.31
b2000_both_s303	geometry:polyhaven/dark	400	76.75	99.12
b2000_both_s303	geometry:polyhaven/light	400	73.75	99.07
b2000_both_s303	palette:aubergine_periwinkle/dark	400	76.50	99.11
b2000_both_s303	palette:aubergine_periwinkle/light	400	74.50	99.05
b2000_both_s303	palette:cobalt_porcelain/dark	400	76.75	99.10
b2000_both_s303	palette:cobalt_porcelain/light	400	72.25	99.02
b2000_both_s303	palette:indigo_lavender/dark	400	75.50	99.07
b2000_both_s303	palette:indigo_lavender/light	400	75.75	99.07
b2000_both_s303	palette:navy_coolrose/dark	400	78.00	99.14

Run	Cell / convention	N	Exact (%)	Square (%)
b2000_both_s303	palette:navy_coolrose/light	400	76.00	99.09
b2000_both_s303	palette:plum_mintgreen/dark	400	79.50	99.19
b2000_both_s303	palette:plum_mintgreen/light	400	77.75	99.13
b2000_both_s303	palette:wine_iceblue/dark	400	78.50	99.13
b2000_both_s303	palette:wine_iceblue/light	400	75.25	99.10
b2000_dark_s304	elevation:08_15/dark	400	16.00	95.70
b2000_dark_s304	elevation:08_15/light	400	0.00	65.31
b2000_dark_s304	elevation:15_25/dark	400	60.00	98.89
b2000_dark_s304	elevation:15_25/light	400	0.00	59.27
b2000_dark_s304	elevation:25_45/dark	400	82.50	99.35
b2000_dark_s304	elevation:25_45/light	400	0.00	55.49
b2000_dark_s304	elevation:45_60/dark	400	79.75	99.36
b2000_dark_s304	elevation:45_60/light	400	0.00	54.27
b2000_dark_s304	framing:compact/dark	400	63.00	98.81
b2000_dark_s304	framing:compact/light	400	0.00	55.77
b2000_dark_s304	framing:standard/dark	400	79.75	99.36
b2000_dark_s304	framing:standard/light	400	0.00	54.27
b2000_dark_s304	geometry:deplorable/dark	400	90.25	99.61
b2000_dark_s304	geometry:deplorable/light	400	0.00	54.13
b2000_dark_s304	geometry:killgorack/dark	400	87.75	99.62
b2000_dark_s304	geometry:killgorack/light	400	0.00	54.34
b2000_dark_s304	geometry:lyricsz/dark	400	79.00	99.45
b2000_dark_s304	geometry:lyricsz/light	400	0.00	55.64
b2000_dark_s304	geometry:nailfighter/dark	400	75.00	99.32
b2000_dark_s304	geometry:nailfighter/light	400	0.00	54.98
b2000_dark_s304	geometry:omie/dark	400	90.75	99.64
b2000_dark_s304	geometry:omie/light	400	0.00	55.39
b2000_dark_s304	geometry:phuong/dark	400	86.50	99.55
b2000_dark_s304	geometry:phuong/light	400	0.00	55.16
b2000_dark_s304	geometry:pjhanzlik/dark	400	76.00	99.32
b2000_dark_s304	geometry:pjhanzlik/light	400	0.00	55.94
b2000_dark_s304	geometry:polyhaven/dark	400	79.75	99.36
b2000_dark_s304	geometry:polyhaven/light	400	0.00	54.27
b2000_dark_s304	palette:aubergine_periwinkle/dark	400	82.00	99.42
b2000_dark_s304	palette:aubergine_periwinkle/light	400	0.00	54.29
b2000_dark_s304	palette:cobalt_porcelain/dark	400	80.25	99.33
b2000_dark_s304	palette:cobalt_porcelain/light	400	0.00	54.84
b2000_dark_s304	palette:indigo_lavender/dark	400	81.50	99.40
b2000_dark_s304	palette:indigo_lavender/light	400	0.00	54.29
b2000_dark_s304	palette:navy_coolrose/dark	400	83.75	99.38
b2000_dark_s304	palette:navy_coolrose/light	400	0.00	54.37
b2000_dark_s304	palette:plum_mintgreen/dark	400	82.75	99.45
b2000_dark_s304	palette:plum_mintgreen/light	400	0.00	54.07
b2000_dark_s304	palette:wine_iceblue/dark	400	80.25	99.39
b2000_dark_s304	palette:wine_iceblue/light	400	0.00	54.10
b2000_both_s304	elevation:08_15/dark	400	16.00	95.64
b2000_both_s304	elevation:08_15/light	400	14.50	95.48
b2000_both_s304	elevation:15_25/dark	400	56.75	98.83
b2000_both_s304	elevation:15_25/light	400	56.25	98.79
b2000_both_s304	elevation:25_45/dark	400	79.25	99.27
b2000_both_s304	elevation:25_45/light	400	78.00	99.18
b2000_both_s304	elevation:45_60/dark	400	76.00	99.09

Run	Cell / convention	N	Exact (%)	Square (%)
b2000_both_s304	elevation:45_60/light	400	73.50	99.00
b2000_both_s304	framing:compact/dark	400	60.75	98.73
b2000_both_s304	framing:compact/light	400	61.25	98.70
b2000_both_s304	framing:standard/dark	400	76.00	99.09
b2000_both_s304	framing:standard/light	400	73.50	99.00
b2000_both_s304	geometry:deplorable/dark	400	89.25	99.44
b2000_both_s304	geometry:deplorable/light	400	88.75	99.40
b2000_both_s304	geometry:killgorack/dark	400	85.00	99.29
b2000_both_s304	geometry:killgorack/light	400	84.25	99.23
b2000_both_s304	geometry:lyricsz/dark	400	73.00	99.08
b2000_both_s304	geometry:lyricsz/light	400	72.25	99.04
b2000_both_s304	geometry:nailfighter/dark	400	72.50	99.08
b2000_both_s304	geometry:nailfighter/light	400	70.00	99.02
b2000_both_s304	geometry:omie/dark	400	91.00	99.48
b2000_both_s304	geometry:omie/light	400	89.50	99.43
b2000_both_s304	geometry:phuong/dark	400	82.00	99.27
b2000_both_s304	geometry:phuong/light	400	82.25	99.24
b2000_both_s304	geometry:pjhanzlik/dark	400	78.00	99.20
b2000_both_s304	geometry:pjhanzlik/light	400	75.00	99.14
b2000_both_s304	geometry:polyhaven/dark	400	76.00	99.09
b2000_both_s304	geometry:polyhaven/light	400	73.50	99.00
b2000_both_s304	palette:aubergine_periwinkle/dark	400	76.75	99.11
b2000_both_s304	palette:aubergine_periwinkle/light	400	74.00	99.02
b2000_both_s304	palette:cobalt_porcelain/dark	400	74.75	99.13
b2000_both_s304	palette:cobalt_porcelain/light	400	75.00	99.05
b2000_both_s304	palette:indigo_lavender/dark	400	75.50	99.12
b2000_both_s304	palette:indigo_lavender/light	400	76.50	99.08
b2000_both_s304	palette:navy_coolrose/dark	400	77.75	99.16
b2000_both_s304	palette:navy_coolrose/light	400	76.00	99.06
b2000_both_s304	palette:plum_mintgreen/dark	400	79.50	99.17
b2000_both_s304	palette:plum_mintgreen/light	400	75.50	99.05
b2000_both_s304	palette:wine_iceblue/dark	400	77.00	99.09
b2000_both_s304	palette:wine_iceblue/light	400	75.25	99.03
b2000_dark_s305	elevation:08_15/dark	400	15.75	95.79
b2000_dark_s305	elevation:08_15/light	400	0.00	66.46
b2000_dark_s305	elevation:15_25/dark	400	64.25	98.98
b2000_dark_s305	elevation:15_25/light	400	0.00	60.62
b2000_dark_s305	elevation:25_45/dark	400	84.75	99.45
b2000_dark_s305	elevation:25_45/light	400	0.00	56.79
b2000_dark_s305	elevation:45_60/dark	400	81.25	99.42
b2000_dark_s305	elevation:45_60/light	400	0.00	55.46
b2000_dark_s305	framing:compact/dark	400	65.25	99.05
b2000_dark_s305	framing:compact/light	400	0.00	56.34
b2000_dark_s305	framing:standard/dark	400	81.25	99.42
b2000_dark_s305	framing:standard/light	400	0.00	55.46
b2000_dark_s305	geometry:deplorable/dark	400	91.25	99.69
b2000_dark_s305	geometry:deplorable/light	400	0.00	54.80
b2000_dark_s305	geometry:killgorack/dark	400	89.25	99.64
b2000_dark_s305	geometry:killgorack/light	400	0.00	55.39
b2000_dark_s305	geometry:lyricsz/dark	400	74.50	99.39
b2000_dark_s305	geometry:lyricsz/light	400	0.00	56.94
b2000_dark_s305	geometry:nailfighter/dark	400	75.00	99.35

Run	Cell / convention	N	Exact (%)	Square (%)
b2000_dark_s305	geometry:nailfighter/light	400	0.00	56.01
b2000_dark_s305	geometry:omie/dark	400	91.50	99.67
b2000_dark_s305	geometry:omie/light	400	0.00	56.06
b2000_dark_s305	geometry:phuong/dark	400	87.75	99.57
b2000_dark_s305	geometry:phuong/light	400	0.00	56.08
b2000_dark_s305	geometry:pjhanzlik/dark	400	75.00	99.32
b2000_dark_s305	geometry:pjhanzlik/light	400	0.00	56.82
b2000_dark_s305	geometry:polyhaven/dark	400	81.25	99.42
b2000_dark_s305	geometry:polyhaven/light	400	0.00	55.46
b2000_dark_s305	palette:aubergine_periwinkle/dark	400	82.75	99.45
b2000_dark_s305	palette:aubergine_periwinkle/light	400	0.00	55.29
b2000_dark_s305	palette:cobalt_porcelain/dark	400	80.25	99.39
b2000_dark_s305	palette:cobalt_porcelain/light	400	0.00	55.83
b2000_dark_s305	palette:indigo_lavender/dark	400	83.00	99.47
b2000_dark_s305	palette:indigo_lavender/light	400	0.00	55.20
b2000_dark_s305	palette:navy_coolrose/dark	400	84.50	99.47
b2000_dark_s305	palette:navy_coolrose/light	400	0.00	55.64
b2000_dark_s305	palette:plum_mintgreen/dark	400	84.50	99.47
b2000_dark_s305	palette:plum_mintgreen/light	400	0.00	54.97
b2000_dark_s305	palette:wine_iceblue/dark	400	83.00	99.45
b2000_dark_s305	palette:wine_iceblue/light	400	0.00	54.99
b2000_both_s305	elevation:08_15/dark	400	13.25	95.21
b2000_both_s305	elevation:08_15/light	400	13.00	95.11
b2000_both_s305	elevation:15_25/dark	400	60.25	98.83
b2000_both_s305	elevation:15_25/light	400	58.25	98.77
b2000_both_s305	elevation:25_45/dark	400	83.00	99.28
b2000_both_s305	elevation:25_45/light	400	79.00	99.14
b2000_both_s305	elevation:45_60/dark	400	79.50	99.21
b2000_both_s305	elevation:45_60/light	400	79.50	99.14
b2000_both_s305	framing:compact/dark	400	65.25	98.88
b2000_both_s305	framing:compact/light	400	64.25	98.80
b2000_both_s305	framing:standard/dark	400	79.50	99.21
b2000_both_s305	framing:standard/light	400	79.50	99.14
b2000_both_s305	geometry:deplorable/dark	400	91.00	99.46
b2000_both_s305	geometry:deplorable/light	400	89.50	99.41
b2000_both_s305	geometry:killgorack/dark	400	86.50	99.34
b2000_both_s305	geometry:killgorack/light	400	87.00	99.30
b2000_both_s305	geometry:lyricsz/dark	400	72.50	99.05
b2000_both_s305	geometry:lyricsz/light	400	74.25	99.04
b2000_both_s305	geometry:nailfighter/dark	400	71.00	99.00
b2000_both_s305	geometry:nailfighter/light	400	72.25	99.00
b2000_both_s305	geometry:omie/dark	400	91.00	99.48
b2000_both_s305	geometry:omie/light	400	88.75	99.38
b2000_both_s305	geometry:phuong/dark	400	84.75	99.33
b2000_both_s305	geometry:phuong/light	400	82.75	99.18
b2000_both_s305	geometry:pjhanzlik/dark	400	81.75	99.25
b2000_both_s305	geometry:pjhanzlik/light	400	79.50	99.16
b2000_both_s305	geometry:polyhaven/dark	400	79.50	99.21
b2000_both_s305	geometry:polyhaven/light	400	79.50	99.14
b2000_both_s305	palette:aubergine_periwinkle/dark	400	84.25	99.28
b2000_both_s305	palette:aubergine_periwinkle/light	400	82.50	99.18
b2000_both_s305	palette:cobalt_porcelain/dark	400	80.25	99.21

Run	Cell / convention	N	Exact (%)	Square (%)
b2000_both_s305	palette:cobalt_porcelain/light	400	80.25	99.17
b2000_both_s305	palette:indigo_lavender/dark	400	81.75	99.25
b2000_both_s305	palette:indigo_lavender/light	400	81.75	99.20
b2000_both_s305	palette:navy_coolrose/dark	400	81.75	99.23
b2000_both_s305	palette:navy_coolrose/light	400	82.75	99.20
b2000_both_s305	palette:plum_mintgreen/dark	400	84.25	99.31
b2000_both_s305	palette:plum_mintgreen/light	400	83.00	99.21
b2000_both_s305	palette:wine_iceblue/dark	400	81.50	99.22
b2000_both_s305	palette:wine_iceblue/light	400	80.75	99.16
b8000_dark_s303	elevation:08_15/dark	400	14.25	95.77
b8000_dark_s303	elevation:08_15/light	400	0.00	68.02
b8000_dark_s303	elevation:15_25/dark	400	62.50	98.89
b8000_dark_s303	elevation:15_25/light	400	0.00	61.00
b8000_dark_s303	elevation:25_45/dark	400	85.00	99.37
b8000_dark_s303	elevation:25_45/light	400	0.00	57.26
b8000_dark_s303	elevation:45_60/dark	400	86.00	99.36
b8000_dark_s303	elevation:45_60/light	400	0.00	55.82
b8000_dark_s303	framing:compact/dark	400	69.00	98.97
b8000_dark_s303	framing:compact/light	400	0.00	56.94
b8000_dark_s303	framing:standard/dark	400	86.00	99.36
b8000_dark_s303	framing:standard/light	400	0.00	55.82
b8000_dark_s303	geometry:deplorable/dark	400	89.00	99.44
b8000_dark_s303	geometry:deplorable/light	400	0.00	55.78
b8000_dark_s303	geometry:killgorack/dark	400	88.25	99.39
b8000_dark_s303	geometry:killgorack/light	400	0.00	55.70
b8000_dark_s303	geometry:lyricsz/dark	400	78.75	99.11
b8000_dark_s303	geometry:lyricsz/light	400	0.00	57.60
b8000_dark_s303	geometry:nailfighter/dark	400	75.75	99.18
b8000_dark_s303	geometry:nailfighter/light	400	0.00	56.63
b8000_dark_s303	geometry:omie/dark	400	90.00	99.44
b8000_dark_s303	geometry:omie/light	400	0.00	56.89
b8000_dark_s303	geometry:phuong/dark	400	87.25	99.30
b8000_dark_s303	geometry:phuong/light	400	0.00	56.52
b8000_dark_s303	geometry:pjhanzlik/dark	400	82.75	99.28
b8000_dark_s303	geometry:pjhanzlik/light	400	0.00	57.40
b8000_dark_s303	geometry:polyhaven/dark	400	86.00	99.36
b8000_dark_s303	geometry:polyhaven/light	400	0.00	55.82
b8000_dark_s303	palette:aubergine_periwinkle/dark	400	85.50	99.34
b8000_dark_s303	palette:aubergine_periwinkle/light	400	0.00	55.79
b8000_dark_s303	palette:cobalt_porcelain/dark	400	86.25	99.36
b8000_dark_s303	palette:cobalt_porcelain/light	400	0.00	56.31
b8000_dark_s303	palette:indigo_lavender/dark	400	85.50	99.34
b8000_dark_s303	palette:indigo_lavender/light	400	0.00	55.80
b8000_dark_s303	palette:navy_coolrose/dark	400	86.50	99.36
b8000_dark_s303	palette:navy_coolrose/light	400	0.00	56.00
b8000_dark_s303	palette:plum_mintgreen/dark	400	86.50	99.39
b8000_dark_s303	palette:plum_mintgreen/light	400	0.00	55.37
b8000_dark_s303	palette:wine_iceblue/dark	400	83.25	99.32
b8000_dark_s303	palette:wine_iceblue/light	400	0.00	55.49
b8000_both_s303	elevation:08_15/dark	400	14.50	95.84
b8000_both_s303	elevation:08_15/light	400	14.75	95.71
b8000_both_s303	elevation:15_25/dark	400	62.75	99.05

Run	Cell / convention	N	Exact (%)	Square (%)
b8000_both_s303	elevation:15_25/light	400	61.25	98.89
b8000_both_s303	elevation:25_45/dark	400	84.25	99.46
b8000_both_s303	elevation:25_45/light	400	81.25	99.32
b8000_both_s303	elevation:45_60/dark	400	86.50	99.49
b8000_both_s303	elevation:45_60/light	400	85.25	99.43
b8000_both_s303	framing:compact/dark	400	65.00	98.96
b8000_both_s303	framing:compact/light	400	65.00	98.91
b8000_both_s303	framing:standard/dark	400	86.50	99.49
b8000_both_s303	framing:standard/light	400	85.25	99.43
b8000_both_s303	geometry:deplorable/dark	400	93.75	99.61
b8000_both_s303	geometry:deplorable/light	400	90.75	99.49
b8000_both_s303	geometry:killgorack/dark	400	90.75	99.54
b8000_both_s303	geometry:killgorack/light	400	86.50	99.45
b8000_both_s303	geometry:lyricsz/dark	400	81.25	99.39
b8000_both_s303	geometry:lyricsz/light	400	79.50	99.28
b8000_both_s303	geometry:nailfighter/dark	400	75.50	99.32
b8000_both_s303	geometry:nailfighter/light	400	74.25	99.21
b8000_both_s303	geometry:omie/dark	400	91.00	99.58
b8000_both_s303	geometry:omie/light	400	91.25	99.54
b8000_both_s303	geometry:phuong/dark	400	89.00	99.54
b8000_both_s303	geometry:phuong/light	400	90.00	99.46
b8000_both_s303	geometry:pjhanzlik/dark	400	82.00	99.42
b8000_both_s303	geometry:pjhanzlik/light	400	79.25	99.32
b8000_both_s303	geometry:polyhaven/dark	400	86.50	99.49
b8000_both_s303	geometry:polyhaven/light	400	85.25	99.43
b8000_both_s303	palette:aubergine_periwinkle/dark	400	87.00	99.48
b8000_both_s303	palette:aubergine_periwinkle/light	400	85.50	99.40
b8000_both_s303	palette:cobalt_porcelain/dark	400	86.00	99.48
b8000_both_s303	palette:cobalt_porcelain/light	400	84.00	99.39
b8000_both_s303	palette:indigo_lavender/dark	400	87.50	99.49
b8000_both_s303	palette:indigo_lavender/light	400	85.00	99.41
b8000_both_s303	palette:navy_coolrose/dark	400	88.00	99.51
b8000_both_s303	palette:navy_coolrose/light	400	85.25	99.41
b8000_both_s303	palette:plum_mintgreen/dark	400	88.00	99.52
b8000_both_s303	palette:plum_mintgreen/light	400	85.50	99.41
b8000_both_s303	palette:wine_iceblue/dark	400	87.25	99.50
b8000_both_s303	palette:wine_iceblue/light	400	85.75	99.41

B Preregistered protocols

The following files are sanitized export copies of the preregistrations and amendments. Their bytes differ from private operational originals only through the documented text-only export policy; the evidence index records the hash of each copy. They are included verbatim from the copied evidence tree rather than reconstructed from the results. Some source documents append execution records after their original preregistration. These later entries are not pre-experiment predictions; the recorded preregistration and source commits identify the earlier protocol versions. Commit identifiers in these documents refer to the author's private development history and do not resolve in the public repository.

BENCHMARK_V2_2026-09-10.md

```
# C1 preregistration: paired convention-controlled benchmark v2

Registered before generation/rendering. This benchmark is a diagnostic research
artifact, not a claim of model superiority. Historical results remain unchanged.

## Fixed design

- 500 fresh legal position groups, one position per independently sampled game.
  Shared across all factor panels. `position_group` is SHA-256 of canonical
  piece-placement FEN. A seeded permutation assigns 400 groups to `dev` and 100
  to `final_test`; every view of a group remains in its partition. Seed 20260910.
- Four elevation bands: 8-15, 15-25, 25-45, 45-60 degrees. Seven geometry families:
  Poly Haven, Deplorable, KillGorack, Lyricsz, Omie, Phuong, PJ Hanzlik. Six held-out
  palettes. Two framing bands: compact 0.30-0.40 and standard 0.52-0.61 board width.
- Every factor level appears at both parity levels: `a1_dark` reference and
  `a1_light` synthetic convention stress. 19 factor levels × 2 × 500 = **19,000
  cell-images**, comprising **15,200 dev and 3,800 final-test cell-images**.
- Reference: Poly Haven, elevation 45-60, standard framing, approved palette 0
  (`copper_pale_teal`), board light RGB [215,206,190], dark RGB [170,148,126].
  The shared reference geometry/elevation/framing views supply paired reference
  images for the six palette treatments. Identical reference views may occur in
  several panels; report duplicate image hashes, not fictitious independent data.
- Within each group, share azimuth, normalized elevation/framing draws, roll and
  position. Only the varied factor changes; elevation/framing require the
  corresponding physical camera parameters to change. Other panels use reference
  settings. Render-derived distance/projection metadata are recorded, not treated
  as independently held-fixed quantities.
- Nailfighter may be added only if its renderer and source-licence validation
  together take less than one hour, without changing any existing group/partition.
  Inclusion is decided before full rendering, not from model results. It adds
  1,000 cell-images (800 dev / 200 final), for totals 20,000 / 16,000 / 4,000.
  Otherwise report it deferred. No additional rendering-look or occlusion axes.

## Exclusions and lineage

Reject canonical exact and all eight D4 equivalents against every inventoried
training/public/reference pool, historical benchmark and prior experiment cohort,
including the complete 6,100,960-row Lichess source. Deduplicate paired views to
position groups for audits. Record source row counts, distinct counts, content
hashes, `excluded_fen_count`, `excluded_fen_sha256`, and policy. No silently omitted
or empty required source. Cross-partition and later adaptation-pool overlap is
forbidden. Material-only similarities are descriptive, not rejection conditions.

Use one explicit parity switch in each renderer lineage, keeping pinned geometry
identity and legacy guard behavior. New dataset roles are explicit and separate
from historical roles. Produce manifests before rendering, then commit the
implementation and render from a fresh clone on render-host. Record the actual checkout
commit, code-byte hashes, external asset hashes and CPU browser/backend identity.
No copied source edits after source freeze. External assets are separately pinned.
```

```
## Final-test firewall and validation

`final_test` is never used for recognition, localization-detector scoring,
attention inspection, selection, timing or training before the designated final
evaluation. Rendering and integrity/GT projection checks are permitted. The
schema-v2 freeze receipt includes the complete final-test image-hash list. Scorers
must reject final-test requests by default; this task provides no final-scoring
override. Freeze generation does not call a detector, even for development.

Validate paired labels/camera/factors, geometry guards, independent asymmetric
corner fixtures, all images/labels/hashes, exact/D4 separation and expected cell
counts. Run rendering and tool tests on render-host, classify baseline failures by exact
test identity, and reject new regressions. Known failure counts are not a waiver
for unrelated new failures. Completion requires an artifact-hashed schema-v2
receipt and complete exclusion audit. `release_ready` remains false pending the
separate licence review; no public upload or model result is part of C1.

## Execution record (append after completion)

### Inclusion decision before rendering

Nailfighter is included. Its existing converted mesh was reusable; renderer
support and validation completed inside the one-hour limit. The author's asset
page explicitly identifies the licence as **Creative Commons Zero v1.0 Universal**:
[LowPoly Chess Set](https://nailfighter.itch.io/lowpoly-chess-set), checked
2026-09-10 UTC. The converted asset digest is
`0d96781041e87b0330f7646cb0720f63acbb42a8f6806d935d81dace20a6951a`.
This is evidence for this asset's inclusion, not completion of the overall
release licence review. The locked C1 totals are therefore **20,000 cell-images,
16,000 dev and 4,000 final_test**, across 500 independent position groups.

The adaptation pool has eight geometry levels (1,000 groups each); its other
balanced marginals and 16,000-image total are unchanged. Palette levels denote
complete supplied material schemes, including RGB, roughness and metalness-not
an RGB-only intervention. Final-test performance remains sealed.

All execution is on render-host; rendering uses CPU SwiftShader from a script-file-backed
named tmux session. Preserve other jobs and retain incomplete artifacts on failure.
The final entry will give the locked cell table, source commit, exclusion counts,
receipt hashes, exact rerender command and test evidence. Emit a one-line status
when the C1 completion receipt lands. No model scoring is authorized by this task.

## C1 completion - 2026-09-10 UTC
Completed at **04:58:28 UTC** from clean source commit
`01c52fb9a69a5b36759b310ae18d26dd8ce662c1`. Renderer smoke fixtures covered all
eight geometries at high/low elevation and both parities (32 images), outside
the benchmark populations. No detector or recognizer was invoked on C1.
| Factor | Level | Dark dev | Dark final | Light dev | Light final |
|---|---|---:|---:|---:|---:|
| elevation | 08_15 | 400 | 100 | 400 | 100 |
| elevation | 15_25 | 400 | 100 | 400 | 100 |
| elevation | 25_45 | 400 | 100 | 400 | 100 |
| elevation | 45_60 | 400 | 100 | 400 | 100 |
| framing | compact | 400 | 100 | 400 | 100 |
| framing | standard | 400 | 100 | 400 | 100 |
| geometry | deplorable | 400 | 100 | 400 | 100 |
| geometry | killgorack | 400 | 100 | 400 | 100 |
| geometry | lyricsz | 400 | 100 | 400 | 100 |
| geometry | nailfighter | 400 | 100 | 400 | 100 |
| geometry | omie | 400 | 100 | 400 | 100 |
| geometry | phuong | 400 | 100 | 400 | 100 |
| geometry | pjhanzlik | 400 | 100 | 400 | 100 |
| geometry | polyhaven | 400 | 100 | 400 | 100 |
| palette | aubergine_periwinkle | 400 | 100 | 400 | 100 |
| palette | cobalt_porcelain | 400 | 100 | 400 | 100 |
| palette | indigo_lavender | 400 | 100 | 400 | 100 |
| palette | navy_coolrose | 400 | 100 | 400 | 100 |
| palette | plum_mintgreen | 400 | 100 | 400 | 100 |
```

```
| palette | wine_iceblue | 400 | 100 | 400 | 100 |

There are **20,000 cell-images but 18,000 unique PNG hashes**, as expected:
the three reference cells deliberately reuse identical FEN/camera/material
configurations. The independent sampling units are **500 position groups**, not
20,000 unrelated observations. Final-test performance remains unopened.

Exclusion evidence: **67 named source pools**, **6,199,823 distinct excluded
placements** for C1; the separate adaptation pool excludes those plus all 500 C1
positions (**6,200,323**). The source inventory includes the verified 6,100,960-row
pre-move Lichess snapshot. The combined 8,500 generated position groups are
exact/D4-unique and have **zero exact/D4 leakage**. Material-only matches (8,077)
are descriptive, not positional leakage or a rejection criterion.

- C1 receipt SHA-256: `[redacted: withheld digest]`.
- Exclusion basis SHA-256: `[redacted: withheld digest]`.
- The complete final-test image-hash list is inside the C1 receipt.
- Receipts and cell summaries: [`benchmark_v2_2026-09-10/`](../tools/reports/paper/benchmark_v2_2026-09-10/).
- `release_ready=false`: this is not a publication or completed licence review.

### Rerender command and software boundary

Use the clean source commit above and the frozen design manifests/assets. Set
`RUN` to the preserved run directory, `REPO` to that clean checkout,
`PLAYWRIGHT_MODULE` to the pinned Playwright entry point, and `CHROME_BIN` to the
Chromium executable for the recorded build. The standalone rerender command (the same committed entry point
↪  used by the launcher) is:

```sh
CUDA_VISIBLE_DEVICES="" PYTHONDONTWRITEBYTECODE=1 OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 \
python tools/paper/run_benchmark_v2.py \
  --run "$RUN" --repo "$REPO" --cohort benchmark \
  --playwright "$PLAYWRIGHT_MODULE" --browser "$CHROME_BIN" --workers 4
```

The runner refuses existing output shards. A fresh rerender needs a separate run
directory with the same preserved design files and no existing `renders/` outputs;
it must not overwrite this freeze. Rendering uses Node v24.15.0, Chromium
149.0.7827.55, Three.js r128 and the verified CPU SwiftShader backend. Full
command scripts and private operational inventories remain with the run on render-host;
only the release allowlist should be exported, not historical repository content.

Test and independent-review evidence

The fresh render-host source run produced **472 passed, 17 failed, 3 skipped**. All failure
identities, including the three failing Blender subtests, were present in the
recorded baseline; there were **zero new failure identities**. The old "2+7"
baseline count was incomplete, so this is not described as a globally green suite.
The transfer-test import repair removed all five recorded transfer failures;
execution-only image/checkpoint dependencies are now lazy-loaded.

Independent metadata review verified all source/asset and shard-manifest hashes,
paired cameras, partition counts, adaptation marginals/nested budgets, exact/D4
separation and exclusion-audit bindings. It did not inspect final-test images or
run performance scoring. The freeze validated every PNG/label, analytic corner
projection and parity pair. Focused CPU tests and static checks passed.

ADAPTATION_POOL_2026-09-10.md

C2 data preregistration: separate matched-position adaptation pool

Registered before generation/rendering. This is the data prerequisite for the
later encoder-LoRA experiment, not authorization to begin training or to score
the sealed final partition. The complete training recipe is preregistered later.

Population and exclusion
```

Generate **8,000 fresh position groups**, one position per independently sampled
legal game; render one scene per group at both `a1_dark` and `a1_light`, producing
16,000 images. All views are `partition=adaptation_train`. Generation seed
20260911; C1 generation/partition seed remains 20260910.

Before accepting any position, reject canonical exact and D4 overlap against all
C1 groups (dev and final_test), historical benchmark positions, all prior cohorts,
every inventoried training/public/reference pool and the complete Lichess corpus.
Publish count/hash evidence, not an unsupported blanket cleanliness claim. No
source may be silently omitted. The generator's own accepted-position set also
rejects duplicates and D4 equivalents within this pool.

Scene sampling and nested budgets

Sample uniformly from the supported C1 factor **levels**, using independently
shuffled balanced marginal schedules: four elevation bands, every included
geometry family, reference palette plus the six palette treatments, and both
framing bands. Per-level counts differ by at most one across 8,000 groups. This
allows combinations of known factor levels, not additional factor levels. It
does not sample one-factor panel IDs uniformly, which would heavily overweight
the reference geometry and camera settings.

Draw nuisance camera parameters independently per group; both parity twins share
the identical scene and camera. Parity is the only within-group intervention.
The pool uses the same pinned renderer/source/asset lineage as C1.

A single permutation with seed **20260912** defines nested prefixes of **500,
2,000 and 8,000 position groups**, fixed before any training or evaluation.
Both arms use the same groups: dark-only uses N distinct images, parity-augmented
uses 2N distinct images, equally divided between parities. These are explicitly
matched-position budgets, not equal distinct-image counts. The later fixed
optimizer-step/batch-size recipe must report exposure and compute separately;
do not disguise duplicated images as a larger data budget.

Evaluation boundary and completion

Future training consumes this pool only; selection/evaluation uses C1 dev, which
is group-disjoint by construction. C1 final_test is never opened for performance
measurement, fitting, timing or selection. No pool image is included in dev
evaluation merely because it originated from the same rendering vocabulary.

Commit this protocol and the implementation before rendering from a clean clone
on render-host. Validate all 16,000 labels, dimensions, hashes, parity twins, group counts,
marginal counts, nested membership and exclusions. The pool receives its own
artifact-hashed completion receipt, with the C1 receipt bound as a prerequisite.
Keep `release_ready=false` pending licence review. Emit a one-line status when
the pool receipt lands. Stop on missing sources, unexpected counts or new test
regressions rather than relaxing the specification.

Completion - 2026-09-10 UTC

Completed at **05:12:48 UTC**, from clean rendering source commit
`01c52fb9a69a5b36759b310ae18d26dd8ce662c1`.

- **8,000 position groups; 16,000 distinct PNG hashes**, with exactly one image
  at each parity per group. No final-test rows occur in this training pool.
- Marginals: each elevation band 2,000 groups; each of eight geometry families
  1,000; each framing 4,000; each of seven palette schemes 1,142 or 1,143.
- The 500 / 2,000 / 8,000 budgets are the exact nested prefixes of the registered
  permutation. Small prefixes are randomly composed, not exactly balanced;
  record their realized composition in the adaptation reports.
- **6,200,323** distinct excluded placements, including every C1 group; exclusion
  basis SHA-256 `[redacted: withheld digest]`.
- Combined C1 + adaptation audit: 8,500 exact/D4-unique groups, zero positional
  leakage against all 67 inventoried sources.
- Pool receipt SHA-256:
  `[redacted: withheld digest]`.

```
  It binds C1 receipt
  `[redacted: withheld digest]`.

Independent streaming verification checked all 16,000 PNG hashes/dimensions,
label hashes, corner hashes, exact path inventories and parity pairs, with zero
mismatches. No image decoding, viewing or model/detector scoring was used by
that verification. The entire render/freeze chain exited zero.

Use the same standalone rerender command documented for C1, substituting
`--cohort adaptation`, in a fresh output directory carrying the preserved design
and manifests. The renderer version, asset hashes and source revision are the
same as C1. Summary and receipts are in
[`benchmark_v2_2026-09-10/`](../tools/reports/paper/benchmark_v2_2026-09-10/).
Dataset pixels, full manifests and private operational inputs remain on render-host.
`release_ready=false`; this data receipt does not authorize publication or
constitute an adaptation-model result.
```

LORA_CURVE_2026-09-10.md

```
# C2 preregistration: upstream LoRA hyperparameters, matched-position budgets

Registered before adaptation training. The later instruction to use upstream
defaults with only the stated protocol changes is controlling: **the head,
queries, decoder and norm are frozen**, superseding the earlier trainable-head
proposal. No extra head-tuning arm is added. The independent unchanged-release
baseline is registered in `C2_BASELINE_2026-09-10.md`.

## Source and parameterization

Pin ChessQueries commit `2a9f3c00619133541c65de4315149037b5dece80` and release
checkpoint SHA-256
`6151bdd98fbe25f32080c097eba7ae75808b5615160e4867240931085fc762e5`.
Use the authors' LoRA implementation, not a newly written approximation:
[LoRA module](https://github.com/JSeytre/chessqueries/blob/2a9f3c00619133541c65de4315149037b5dece80/chessquer
↪  ies/models/lora.py),
[adaptation configuration](https://github.com/JSeytre/chessqueries/blob/2a9f3c00619133541c65de4315149037b5dec
↪  e80/chessqueries/models/adapt.py),
[few-shot CLI](https://github.com/JSeytre/chessqueries/blob/2a9f3c00619133541c65de4315149037b5dece80/scripts/
↪  lora_fewshot.py).

| Setting | Locked value |
|---|---|
| Trainable weights | Encoder LoRA A/B only; base encoder and entire readout frozen |
| Targets | `attn.qkv`, `attn.proj`, `mlp.fc1`, `mlp.fc2` |
| Rank / alpha / dropout | 8 / 16 / 0.05 (few-shot CLI dropout, not the lower-level API's zero default) |
| Expected trainables | 96 adapted linear modules, 192 tensors, 3,145,728 parameters |
| Optimizer | AdamW, LR 0.0001, weight decay 0, betas (0.9, 0.999), epsilon 1e-8 |
| Scheduler / warm-up | None / zero |
| Training | Batch 6, bf16 autocast, exactly 1,500 optimizer updates |
| Loss | Mean 13-class cross-entropy over all 64 squares |
| Input transform | Authors' full-frame `eval_transform(644)`; no input augmentation or TTA |
| Sampling | Seeded epoch shuffle, drop incomplete batches; 9,000 images presented per run |
| Checkpoint | Fixed final update 1,500; no early stopping or best-on-dev checkpoint selection |
| Evaluation | Batch 4, fp32/TF32, argmax, evaluation mode |

The LR/weight-decay/no-scheduler values come from the upstream
[few-shot implementation](https://github.com/JSeytre/chessqueries/blob/2a9f3c00619133541c65de4315149037b5dece
↪  80/chessqueries/train/fewshot.py)
and adaptation defaults, not a learning rate printed in the README. This uses
upstream **adapter/optimizer hyperparameters**, not an unchanged upstream
experiment: the matched-position data, fixed 1,500 steps, batch/precision, two
arms and explicitly disabled input augmentation are this protocol. Fixed-step
evaluation replaces upstream early stopping/checkpoint selection. Do not hide
these differences behind an unqualified "identical upstream experiment" claim.

## Data and run matrix
```

Training uses only the completed, separate adaptation pool: 8,000 fresh position groups with both parities, exact/D4-disjoint from the entire C1 benchmark and all inventoried historical sources. Its receipt must bind the exact current C1 and design receipts. The seed-20260912 permutation provides nested group prefixes.

Groups	Dark-only images	Both-parity images	Seeds per arm
500	500	1,000	303
2,000	2,000	4,000	303, 304, 305
8,000	8,000	16,000	303

Ten runs, each initialized independently from the same release checkpoint; larger budgets do not continue training from smaller budgets. Seed before LoRA initialization and sampling. The arms match positions and optimizer exposure, **not distinct-image count**: both-parity uses twice as many rendered views, with 9,000 presentations in either arm. Report this cost and realized factor counts at every budget; the smaller random prefixes need not be exactly balanced.

Inject LoRA on CPU before moving the complete model to GPU, then rebuild the optimizer parameter references. Frozen decoder weights must remain in the autograd path to encoder adapters. Verify tensor counts, device/dtype, initial base equivalence, and finite loss/gradients. Do not silently detach the encoder, reduce batch size/resolution, alter the recipe, or resume from a different run.

Evaluation and gates

Evaluate every final checkpoint on all **16,000 C1 dev cell-images**, reporting exact-board and per-square scores per factor-cell × parity and pooled summaries. Never read final-test images or compute final-test performance. Integrity and disjointness checks may read the sealed metadata, as in C1.

Compare with the single hash-bound release baseline. Paired bootstrap uses 2,000 resamples, seed **20260913**, with each position group kept intact across the whole matrix; do not split a group's geometry views into independent strata. Report cell/parity intervals and exact gains/losses, every seed, and failures.

ChessReD test is the public retention check: 2,129 images through the authors' loader, annotation hash and 10,800-image source inventory checked before/after. Require **at least 99.12% exact**, within 0.5 points of the 99.62% reference. This is a retention threshold, not an untouched external-test or superiority claim. A receipt's `accepted` flag denotes this retention gate only; development gains or losses must still be reported. No new minimum dev-gain threshold is invented after seeing results, and there is no test-driven hyperparameter sweep.

Completed negative results receive receipts with `accepted=false`. Execution failures remain incomplete, with progress and available adapter-only snapshots; do not relabel them as successful experiments. Baseline retention failure blocks the adaptation matrix pending a harness/data investigation. Store the final adapter before long evaluation, never overwrite the base checkpoint.

Runtime and stop condition

Run serially on otherwise idle render-host GPU 0, using script-backed named tmux sessions. Check competing GPU processes during execution; never terminate other jobs. Preserve all source/input hashes and artifact-hashed completion receipts, and post one-line statuses when receipts land. Stop after the ten fixed cells; no additional search, selector work, model publication or final-test opening. `release_ready=false` pending the separate licence review. Timing comparisons belong to C4, not these training/evaluation wall times.

LORA_REPORT_PROTOCOL_2026-09-10.md

C2 reporting protocol

Registered while the first fixed adaptation point is still running and before reading any adapted-model evaluation. No change to the training protocol.

```
After the ten-point matrix receipt lands, `tools/paper/lora_report.py` verifies
every artifact-bound point and independently rescores its saved FEN strings.
It verifies complete 400-position-group × 40-view dev coverage, unchanged GT
and public-test membership, source/recipe identity, and the retention flags.
It reads predictions and receipts only, never images or the GPU.

Report all ten seeds/arms/budgets, both parity scores, every factor cell,
ChessReD retention failures, pooled paired changes and gained/lost cell-images.
Keep the preregistered whole-position bootstrap intervals from the bound run
metrics; do not describe this aggregation as an independent CI recomputation.
At 2,000 groups, show mean and sample SD across all three fixed seeds per arm.
No best-seed selection or unregistered significance threshold.

The report must distinguish matched positions/optimizer exposure from distinct
images (N versus 2N), 400 independent groups from 16,000 cell-images, development
scores from untouched final-test/general-real-world claims, and public retention
from overall superiority. No SOTA or zero-cost invariance claim is automatic.

Completion writes Markdown, machine-readable full metrics, and a hashed receipt.
This report gates the post-matrix order: **C2 report → C4 timing → C3 residual**.
`release_ready=false`; failures remain visible and final-test images remain sealed.
```

TIMING_PROTOCOL_2026-09-10.md

```
# C4 preregistration: end-to-end latency on a fixed C1 development subset

Registered before timing and before C2 releases GPU 0. This protocol authorizes
preparation and CPU validation now; it does not authorize concurrent GPU work.
C4 runs immediately after the complete ten-point C2 matrix and before C3.

## Population

Use only C1 `dev`. Seed **20260914** selects 250 `(position_group,
factor_cell)` scenes, balanced across the 20 factor cells as closely as integer
counts allow. Both parity mates are retained, producing exactly **500 images**.
Position groups may repeat across factor cells because C1 has only 400 independent
groups; report unique groups, repeats, and the maximum scenes per group. This is a
latency sample, not 500 independent accuracy observations. No final-test image may
be read or timed, and the CLI exposes no partition or firewall override.

Before model import or CUDA allocation, verify the sealed 20,000-image C1 receipt,
all 16,000 dev image and label hashes, dimensions, partitions, selection receipt,
and source hashes. Recheck them after timing. Require the release-baseline receipt
and all ten completed C2 point receipts, including completed points that failed the
ChessReD retention gate. Every adapter must remain bound to that baseline, the
pinned source/recipe, and `train_head=false`.

## Systems

Run one fresh process per public alias:

- `cq_release`
- `private_full`, `private_raw`, `private_warp_high`, `private_warp_low`
- all ten `lora_{dark|both}_n{500|2000|8000}_s{303|304|305}` registered C2 points

Private aliases do not expose historical member numbers. Private variants are
made with `member_eval.make_variant` in scratch space; weights are symlinked and
never changed. The full system retains its native detector, warp, router, member
mix, rotation TTA, orientation and decoder. Subsets retain the existing
`make_variant` semantics. Consequently these systems do **not** have equal FLOPs,
and C4 must not claim that they do.

The private ensemble is extracted once by the controller from the frozen archive; its size and digests are
↪  [redacted: private artefact]. C4 requires that extraction's receipt and
hashes every extracted source/weight before and after each run; it neither
extracts again nor touches the source archive.
The frozen extraction receipt SHA-256 is [redacted: private artefact];
```

it binds exactly [redacted: private artefact] extracted files plus the separate identity JSON. Extra, missing, or self-consistently re-receipted files are rejected.

Timer

All systems receive the same harness-decoded contiguous uint8 RGB arrays. File decode and BGR-to-RGB conversion are outside the primary timer. Each native pipeline begins from that RGB input. ChessQueries resize, normalization, host-to-device transfer, FP16 forward, argmax, prediction transfer to host and FEN construction remain inside the timer. The private system includes its detector/router/warp/TTA and CPU postprocess.

Use batch size **1** and CUDA FP16 autocast. Run 30 ordered warm-up images, excluded from results, followed by all 500 images in manifest order. Immediately synchronize CUDA before `time.perf_counter()` and after prediction. Save every finite per-image duration and prediction in JSONL, then report median and linearly interpolated p90. Model construction and image decode are excluded. A model-only secondary timer is not required.

Hardware and interference boundary

Run only on render-host with physical GPU 0 exposed as `CUDA_VISIBLE_DEVICES=0`. Record GPU name, driver, total memory, the fixed **270 W** power limit, GPU/memory utilization, Torch/CUDA/Python versions and UTC timestamp. Fail before allocation if another compute PID occupies GPU 0; repeat that check during warm-up/timing and at completion. Never kill another process. C2 owns GPU 0 until its matrix finishes, so C4 must wait rather than probe.

Every output prefix is no-clobber reserved. Successful runs receive artifact-hashed completion receipts and raw timing JSONL. Execution/interference/nonfinite failures retain the incomplete marker and any flushed per-image evidence. These measurements describe this hardware/software/input pipeline only; they are not universal latency or compute-efficiency claims.

The controller first runs `prepare`, then invokes `timing-all` once. `timing-all` launches the 15 aliases serially as fresh child processes and publishes no aggregate completion receipt until all 15 per-model `c4_timing_complete` receipts validate. The aggregate receipt also uses `status=c4_timing_complete`, `release_ready=false`, and `final_test_images_read=false`.

```sh
python tools/paper/timing_harness.py prepare \
  --c1-root "$R" --out "$T/selection"

CUDA_VISIBLE_DEVICES=0 CHESSQUERIES_DATA_ROOT=$HOME/bench_sources \
python tools/paper/timing_harness.py timing-all \
  --c1-root "$R" --selection "$T/selection.jsonl" \
  --matrix-root "$L" --baseline-receipt "$L/results/release.complete.json" \
  --lora-report-receipt "$T/lora_report.complete.json" \
  --cq-repo "$L/upstream" \
  --checkpoint $HOME/bench_experiments/codex_parity_20260909/upstream/checkpoint.safetensors \
  --submission "$T/private_ensemble" \
  --submission-receipt "$T/private_ensemble.complete.json" \
  --out "$T/timing"
```

CPU-only preparation evidence

On 2026-09-10, before C4 authorization, the verified private extraction was copied to fresh scratch space and imported with `CUDA_VISIBLE_DEVICES=""` using the existing render-host runtime. Calling `_load()` initialized all three configured models on CPU, including the DINOv3 member; Torch reported CUDA uninitialized. No image was decoded, no `predict()` call was made, and no inference or timing was performed. The extraction receipt and all [redacted: private artefact] extracted file hashes were identical before and after; scratch was removed. This proves dependency and checkpoint-load compatibility only, not latency, accuracy, or GPU readiness.

RESIDUAL_SHIFT_PROTOCOL_2026-09-10.md

```
# C3 preregistration: residual recognition shift and descriptive attention

This protocol is registered before C3 metrics are computed or attention is
captured. C3 is diagnostic, not a model-selection or mechanism-confirmation
stage. It uses no benchmark-v2 final-test image, score, inspection, timing, or
attention result.

## Frozen population and model output

- Use the completed historical confirmation cohort at
  `$HOME/bench_experiments/codex_transfer_20260910`: 288 position groups,
  six geometry families with 48 groups each, and paired parity renders.
- Analyze exactly the **288 canonical a1-dark** (`condition=parity`) images:
  deplorable, killgorack, lyricsz, omie, phuong and pjhanzlik. Do not analyze the
  byte-identical control here. The historical control renderer assigns the
  light material to its odd `(row + column)` branch; a1 is renderer row 7,
  column 0 and is therefore light. `transfer_cohort.py` creates the `parity`
  renderer by swapping that odd branch to the dark material, so `parity`, not
  `control`, is the a1-dark condition. Before using any rows, require every
  recorded parity-renderer source to equal `swap_parity(control_source)` and
  verify both checkerboard branches explicitly.
- Verify the artifact hashes and bindings in `prepare.complete.json`,
  `render_validation.complete.json` and `confirmation.complete.json`. Rehash
  every selected source image. The prepare manifest, validated confirmation
  inventory and saved predictions must agree by key, FEN and SHA-256.
- Use only rows where `model=default` from the already-saved canonical
  ChessQueries predictions. Selector, rule and adapted-model outputs are not
  C3 inputs. Save a new C3 prediction file containing only those default rows.
- The ground-truth and prediction label order is `.PNBRQKpnbrqk`; all 64
  squares are ordered a8, b8, ..., h8, a7, ..., h1. Pin the release checkpoint
  SHA-256 to
  `6151bdd98fbe25f32080c097eba7ae75808b5615160e4867240931085fc762e5`.

## Locked descriptive metrics

Report overall and separately for all six geometry families:

1. a 13 x 13 count matrix with ground-truth class on rows and predicted class
   on columns;
2. exact 13-class accuracy restricted to ground-truth occupied squares;
3. binary occupied-versus-empty accuracy over all squares, reported separately
   from class accuracy;
4. all-square exact-class accuracy and exact-board accuracy for context; and
5. a 65-bin histogram (0 through 64) of wrong-square counts per board.

These are a characterization of error type and severity. No post-hoc pass
threshold, new selector, corrected prediction, or significance claim is added.
A null, mixed or negative pattern is a valid completed C3 result.

## Outcome-independent attention sample

Before loading the checkpoint, independently select ten canonical images from
each of lyricsz, pjhanzlik and deplorable (30 total). Within a family, rank the
48 records by ascending SHA-256 of
`20260915:group_id:image_sha256`, with record key as a deterministic tiebreak,
and take the first ten. Prediction correctness, confidence and metric outcomes
do not enter selection.

Use the frozen minimal-reproduction model implementation that produced the
saved default predictions; do not substitute the newer public model class or
change the checkpoint architecture/label mapping. Its encoder, query module,
PyTorch decoder, norm and head expose the structural interface consumed by the
visualization helper. Rehash every model/source file recorded by the historical
confirmation receipt before model allocation and after attention capture. Use
the pinned ChessQueries source commit
`2a9f3c00619133541c65de4315149037b5dece80`. Load the same release model and
```

```
authors' `eval_transform(644)` as the saved default predictions. The original
1920 x 1080 frame is squashed to 644 x 644, with no crop or augmentation. Run
at most two boards per attention batch on render-host GPU 0 after C4 and after the LoRA
matrix has released it.

Call `chessqueries.viz.attention.extract_attention(model, batch)` and use
`maps.cross_grid(layer=-1, head=None)`. That upstream method averages heads,
drops CLS/register prefix tokens, and renormalizes patch mass. Preserve all 64
query maps in a compressed raw NPZ in fixed a8-to-h1 order. For every board,
create two 8 x 8 contact sheets containing all 64 fixed query overlays: one
interpolates the heat maps to the 644 x 644 model-input geometry, and one
stretches them to the original 1920 x 1080 frame before thumbnailing. Preserve
those geometries in the sheet tiles: 192 x 192 square tiles for the 644 x 644
view and 192 x 108 16:9 tiles for the source-frame view. Neither boards nor
queries may be chosen after viewing results.

Require finite, nonnegative maps whose patch mass sums to one per query. Set
PyTorch float32 matrix-multiplication precision to `high`, matching the frozen
confirmation run. Check that captured logits match an ordinary forward pass
within `atol=1e-4, rtol=1e-4` (the weight-returning attention path may select a
different numerical kernel), a subsequent ordinary
forward remains unchanged, temporary attention-forward overrides are restored,
and selected argmax predictions reproduce the saved default predictions.
Attention is reported only as descriptive localization evidence. It does not by
itself establish a causal model mechanism, and no forced explanation is a
stopping requirement.

## Outputs and stopping rule

The no-clobber C3 prefix contains default-only predictions, JSON metrics, the
pre-inference selection, raw maps/logits, contact-sheet hashes, and a completion
receipt. The receipt binds all input receipts/manifests/predictions, selected
IDs and images, source code, this protocol, pinned upstream attention source,
and the base checkpoint. Contact-sheet images are transitively bound through an
artifact-hashed manifest. Recheck frozen inputs after attention capture.

Also write a Markdown report with all six family metrics, occupancy-versus-class
error counts and every nonzero wrong-square histogram bin. It links the complete
confusion matrices and attention artifacts and states the mechanism remains
unresolved unless separately supported by an intervention. Display heat colours
are scaled to each query map's own maximum, not absolute cross-query saliency.
Require the upstream checkout to be clean at the pinned commit and its
`chessqueries/viz/attention.py` SHA-256 to be
`af091615c10157259ceddcdb0ce6e0e871c74b046b7ef5b87db0c84939ed9d15`
both before and after capture. Snapshot the C3 runner, protocol and imported
paper-tool dependencies before GPU allocation and require those hashes to remain
unchanged before completing the receipt. Refuse to allocate while another GPU 0
compute PID exists; recheck between batches while allowing only the current PID.

Complete once all 288 default predictions and all 30 x 64 attention maps pass
the locked checks, regardless of whether a coherent residual pattern appears.
An execution failure remains incomplete; use a new output prefix rather than
overwriting it. Do not run C3 until the LoRA report and C4 are complete.
`release_ready=false` remains in force pending the separate licence review.
```

C3_PRECISION_AMENDMENT_2026-09-10.md

```
# C3 R1: precision-only instrumentation correction

Registered before rerunning the full C3 attention sample. The first C3 attempt
at source `027d0c7` remains **failed/incomplete**, and its prefix/log are retained.
The completed C2 matrix, all model/retention results, and C4 are not changed or rerun.

The registered three-condition first-batch diagnostic reproduced the issue:

| Matmul / SDP | Capture maximum absolute logit difference | Original allclose gate | Saved argmax
↪  differences |
```

```
|---|---:|---|---:|
| high / automatic | 0.0018221736 | fail | 0 |
| highest / automatic | 0.0000324249 | pass | 0 |
| highest / math | 0.0000123978 | pass | 0 |

Ordinary-after versus ordinary-before was exactly zero in all three cases;
hooks were restored. This supports a precision/kernel explanation, not a
weight/state mutation. The diagnostic used the same first two already-selected
boards, not a newly selected easy sample. Its first launcher attempt was rejected
before GPU allocation because its log occupied the output prefix; that failure
is retained separately.

**Minimal correction selected by the preregistered priority:** set matmul
precision to `highest` during ordinary-before, upstream attention capture, and
ordinary-after. Keep the automatic SDP backend; forcing math is unnecessary on
this diagnostic. Assert FP32 model/input tensors and restore the caller's matmul
precision even on failure. Log every batch's numerical differences before raising
any validation failure.

All substantive C3 controls are unchanged: the same frozen checkpoint and source,
288 historical canonical rows, identical 30-board selection, all 64 queries,
unchanged **atol=1e-4, rtol=1e-4**, exact saved-prediction argmax agreement,
hook/source/input integrity checks, and no final-test access. A failed exact or
numerical check still stops the run; no further automatic tolerance changes.

Performance/confusion metrics continue to use the original saved default
predictions. The attention instrumentation now uses stricter numerical precision
than their historical TF32-enabled inference; this distinction must accompany
the maps. Attention remains descriptive, with mechanism unresolved.

Execute from a new clean source checkout and new `residual_r1` prefix only after
the original process has exited and GPU 0 is idle. Do not overwrite the original
attempt or relabel its failed gate as a pass. `release_ready=false`.

See [the preregistered diagnostic](C3_NUMERICAL_DIAGNOSTIC_2026-09-10.md) and
[PyTorch's TF32 explanation](https://docs.pytorch.org/docs/2.11/notes/numerical_accuracy.html).
```

C3_NUMERICAL_DIAGNOSTIC_2026-09-10.md

```
# C3 numerical instrumentation diagnostic - registered after the failed first attempt

The original C3 attempt at source `027d0c7` failed its first-batch ordinary versus
captured forward-equivalence check. Its outputs and `residual.incomplete` remain
untouched; no completed C3 receipt exists. C2 and C4 results are unaffected.

Before any diagnostic inference, register this bounded sequence on the **same
first two preselected attention boards**, same checkpoint/source/transform and
batch size. No new boards, learning, metric selection, or final-test access.

1. Reproduce `high` matmul precision with automatic SDP backend.
2. Change only matmul precision to `highest`; retain automatic SDP.
3. Retain `highest` and force the math SDP backend.

For each, record ordinary-before, captured and ordinary-after logits; maximum
absolute differences, original 1e-4 atol/rtol pass flags, hook restoration, and
argmax differences versus the frozen saved predictions. The latter requirement
is exact and unchanged. Weights are not modified. Inputs/sources/selection are
hashed; the probe is isolated on GPU 0 after the completed C4 run.

Ranked hypotheses: TF32 rounding in the non-fused path when requesting weights;
then fused/math reduction differences; finally hook/state mutation if restoration
or equivalence fails even with higher precision/math. This is not a model-quality
or retention test. Do not relax thresholds or choose a new sample to pass.

If a numerical-only setting passes both unchanged gates, preregister the minimal
setting correction for a new C3 attempt/prefix; retain the failed original beside
it. Otherwise report the unresolved instrumentation failure rather than claim
```

```
validated attention. Run no unregistered fourth condition.

Primary implementation context: PyTorch documents that requesting attention
weights disables the optimized SDP path, that backends can differ numerically,
and that TF32 reduces mantissa precision. This motivates the hypotheses; it does
not yet prove the cause of this run's failure.

- [MultiheadAttention, PyTorch
↪  2.11](https://docs.pytorch.org/docs/2.11/generated/torch.nn.MultiheadAttention.html)
- [SDP numerical behavior](https://docs.pytorch.org/docs/2.11/generated/torch.nn.functional.scaled_dot_produc
↪  t_attention.html)
- [TF32 numerical accuracy](https://docs.pytorch.org/docs/2.11/notes/numerical_accuracy.html)
```

C Convention audit

Table 5 records one representative for each of the 56 image sources. It establishes the displayed convention only for the sampled record. In particular, the SLCC sample lacks an independent named-corner anchor and remains unresolved; the table does not support a universal claim that every public image uses a1 dark.

Table 5: One representative per source; unresolved means no independent corner anchor.

Source sample	Named light corners	Labels mirrored?
private train set	a1, h8	No
angles	a1, h8	No
colors_b	a1, h8	No
geometry	a1, h8	No
geometry_b	a1, h8	No
lowangle	a1, h8	No
lowangle_b	a1, h8	No
mixed	a1, h8	No
ph_angles	a1, h8	No
ph_colors	a1, h8	No
ph_colors_glossy	a1, h8	No
ph_geometry	a1, h8	No
ph_geometry_c	a1, h8	No
ph_geometry_d	a1, h8	No
ph_lowangle	a1, h8	No
ph_lowangle_b	a1, h8	No
ph_lowangle_c	a1, h8	No
ph_lowangle_d	a1, h8	No
ph_occlusion_low	a1, h8	No
ph_occlusion_low2	a1, h8	No
ph_staunton_a	a1, h8	No
ph_staunton_b	a1, h8	No
ph_staunton_c	a1, h8	No
ph_staunton_glossy	a1, h8	No
ph_staunton_glossy_low	a1, h8	No
ph_staunton_low	a1, h8	No
cr360_scheme	a8, h1	No
cr360_yaw	a8, h1	No
ph_colors_h	a1, h8	No
ph_colors_h_f	a1, h8	No
ph_geometry_h	a1, h8	No
ph_geometry_h2	a1, h8	No
ph_geometry_h2_f	a1, h8	No
ph_lowangle	a1, h8	No
ph_lowangle_h	a1, h8	No
ph_occlusion_h	a1, h8	No
ChessReD	a8, h1	No
ChessCog	a8, h1	No
SLCC	Unresolved	No
CVChess	a8, h1	No
CR360	a8, h1	No
mixed_data-host	a1, h8	No

Source sample	Named light corners	Labels mirrored?
ph_angles_data-host	a1, h8	No
ph_colors_data-host	a1, h8	No
ph_geometry_data-host	a1, h8	No
ph_lowangle_b_data-host	a1, h8	No
ph_lowangle_c_data-host	a1, h8	No
ph_lowangle_data-host	a1, h8	No
ph_staunton_blender	a1, h8	No
ph_staunton_blender_low	a1, h8	No
ph_staunton_blender_occl	a1, h8	No
ph_staunton_blender_occl2	a1, h8	No
ph_staunton_blender_occl3	a1, h8	No
ph_staunton_blender_occl4	a1, h8	No
ph_staunton_blender_occl4b	a1, h8	No
ph_staunton_blender_shape3	a1, h8	No

D Asset and dataset licence record

The asset table reports the grants observed for the exact frozen geometry lineage. It is an evidence register, not legal advice or a blanket licence for the benchmark, upstream datasets, model weights, or adapters.

Seven geometry families have a recorded CC0 grant. Phuong2647's source page states CC-BY without displaying a version. Under the author decision dated 2026-09-14, the package credits this asset under CC BY 4.0 attribution terms in the family's renders, retaining creator credit, source and licence links, and the conversion and rescaling record. The renders themselves are CC BY-NC 4.0, like every other render in the benchmark. The Phuong render-attribution gate is closed. This is an author-selected package policy, not newly recovered upstream-version evidence. The earlier licence investigations remain historical records; the current decision is in `evidence/docs/records/PHUONG_DECISION_2026-09-14.md`. The asset-licensing status of the benchmark data is separate from the package reproduction check, the publication decision and the withheld model and adapter weights. The receipts do not record the reproduction check or the publication decision.

Table 6: Recorded source grants; Phuong package attribution follows the author decision, not a recovered source version. Renders of every family are CC BY-NC 4.0.

Asset family	Recorded licence	Release qualification
polyhaven	CC0	grant recorded
deplorable	CC0 1.0 Universal	grant recorded
killgorack	CC0	grant recorded
lyricsz	CC0	grant recorded
omie	CC0	grant recorded
phuong	CC-BY; credit Phuong2647	credited under CC BY 4.0
pjhanzlik	CC0	grant recorded
nailfighter	CC0	grant recorded

Table 7: Dataset and position-source register from the recorded licence review.

Dataset / source	Recorded terms	Distribution qualification
Provided private train	No release grant established. Do not export raw pixels/labels, derived caches, samples, or weights trained on it until the rights holder provides a written scope that covers the proposed export.	Deny.
Own v1/v2/adaptation renders	Original work does not erase third-party asset terms. C1/C2 outputs can only enter the conditional visual allowlist when the exact source-family record, hash, and applicable notices travel with the export.	Conditional only.

Dataset / source	Recorded terms	Distribution qualification
ChessReD v2	CC BY-NC-SA 4.0: sharing/adaptation requires attribution, licence link, change indication, non-commercial use, and same-licence treatment for adaptations. The source's own licence does not resolve whether a particular trained model is an adapted work. Terms	Raw/converted photos not in a general release; any licence-compatible non-commercial data export and model claim requires counsel/rights review plus attribution/SA plan.
ChessCog	The OSF API checked for this record reports license: null; the upstream code repository's MIT licence is not a data licence. No permission is inferred from public access.	Deny raw/converted data, samples, and dependent release artifacts pending an explicit dataset grant.
SLCC	The upstream release distinguishes CC BY-NC 4.0 annotations from source broadcasts/reconstructed frames, which remain with their rights holders. Annotation terms do not grant broadcast-frame redistribution.	Deny reconstructed image pixels and image-derived export. Annotation-only reuse remains conditional on exact CC BY-NC compliance and scope review.
CVChess	Repository metadata shows no declared licence and no licence file; no separate image-data grant was found.	Deny raw/converted images and dependent public release artifacts pending explicit author/rights-holder permission.
ChessRender360 / CR360	CC BY 4.0 per the recorded versioned source. Sharing/adaptation needs creator credit, licence link, and changes indication; the record must be rechecked against the exact exported version. CC BY terms	Conditional data/export allow after a versioned record, attribution, change log, and model-lineage review.
Lichess puzzles / 6,100,960-row FEN snapshot	The open-database page expressly releases its exports under CC0 and says they may be downloaded, modified, and redistributed. The source snapshot must still be identified by its exact download/retrieval date and file hash; the live page's current count is not the historical snapshot identity.	Raw snapshot export is conditionally permitted by the stated CC0 grant once the exact file/hash is identified; it is not required for a C1 benchmark-data export. Exclusion hashes/counts may be published only after confirming they disclose no rows or account-linked data.

E Descriptive attention sheets

The compact excerpt in Figure 6 is for visual inspection, not a performance estimate or a causal test. Three query sheets ship in the release package. The full set is not released.

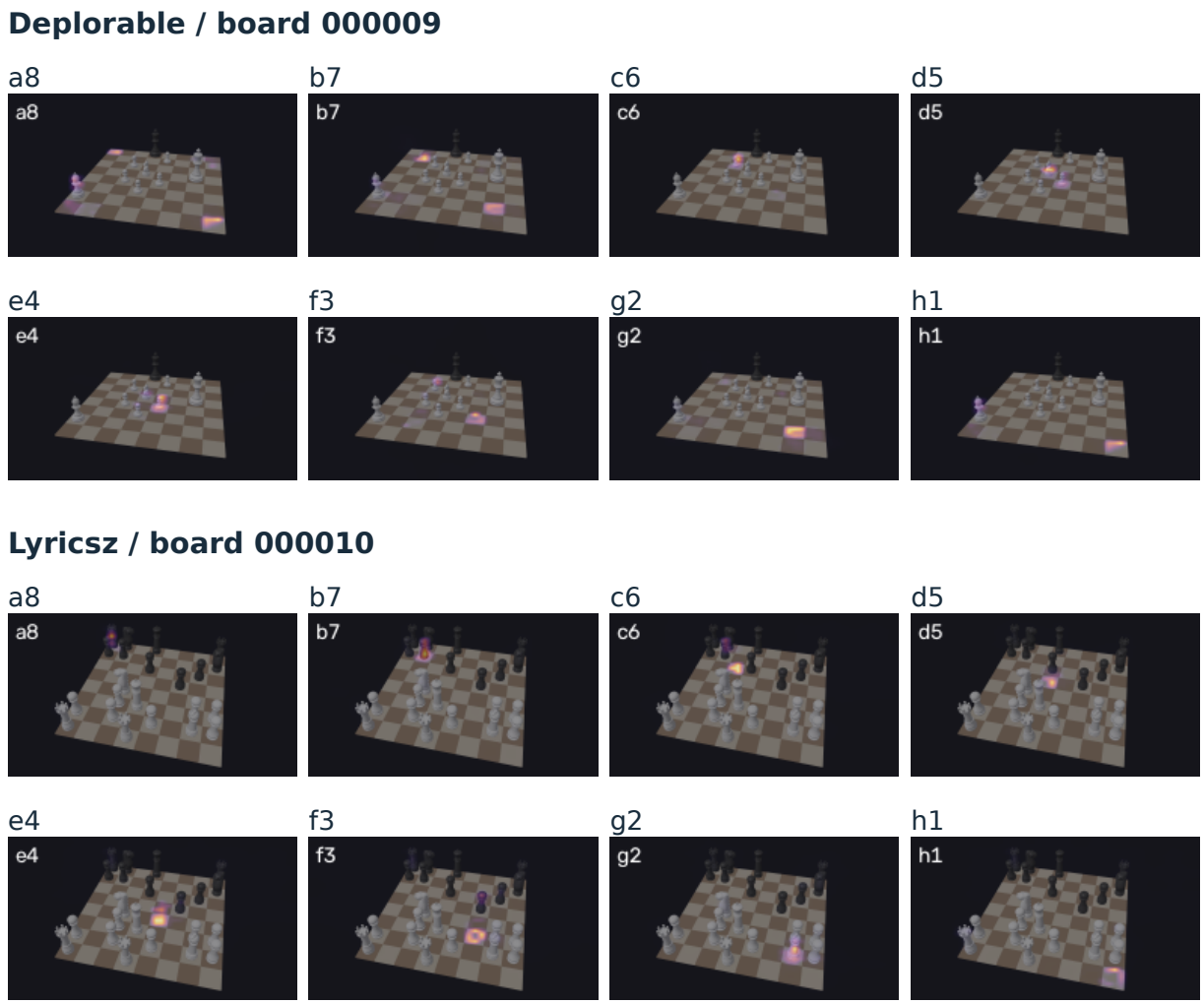


Figure 6: Fixed, outcome-independent excerpt. Select the first two geometry names lexicographically in the frozen attention sample, then the lowest zero-padded board ID in each: Deplorable 000009 and Lyricsz 000010. For both boards use the same eight diagonal queries, a8, b7, c6, d5, e4, f3, g2 and h1. No prediction, correctness, confidence or attention magnitude enters this selection. Tiles are unchanged crops of the recorded overlays, normalized per query, not a shared saliency scale. The full set has 60 sheets (30 boards, 64 queries, two projections). Three sheets ship in the package, in `figures/attention/`. The full set is not released.

F Receipt and test index

The evidence index carries the digest of each copied file. Digests of the unredacted originals are not shipped. Table 8 lists the completion receipts consumed by the generated tables and figures, and the result summaries behind the Section 4 numbers (parity probe, crop diagnostic, baseline and member scores, identifier-level holdout check, real-board check). Per-image prediction files for those runs are not part of this package. A completion receipt records the checks stated by its own schema; it is a checksummed record, not an independent attestation that no unrecorded experiment occurred. It does not close the separate data, model, or asset distribution gates.

Table 8: Receipt and result-summary index. Paths and the hash of each copy are in the evidence index.

Receipt filename	Status / schema
adaptation.complete.json	schema_v2_frozen
benchmark.complete.json	schema_v2_frozen
design.complete.json	design_locked_before_rendering
exclusion_audit.complete.json	all_required_sources_audited
PARITY_CONVENTIONS.complete.json	parity-conventions-audit/v1
PHUONG_LICENCE_RECOVERY.complete.json	licence_recovery_complete_version_unresolved
PHUONG_RELEASE_STATUS_2026-09-14.complete.json	operator_decision_applied
PHUONG_RELEASE_STATUS_2026-09-15.complete.json	author_notice_bindings_updated
PHUONG_RELEASE_STATUS_2026-10-12.complete.json	public_doc_notice_bindings_updated
RELEASE_LICENCE_REVIEW.complete.json	licence_review_complete_with_open_release_gates
lora_report.complete.json	lora_curve_report_complete
result_r1.complete.json	completed_exploratory_probe
precision_results.complete.json	c3_precision_probe_complete
overall_recovery.complete.json	complete_after_c3_precision_instrumentation_recovery
residual_r1.complete.json	c3_residual_complete
timing.complete.json	c4_timing_complete
cq_release.complete.json	c4_timing_complete
lora_both_n2000_s303.complete.json	c4_timing_complete
lora_both_n2000_s304.complete.json	c4_timing_complete
lora_both_n2000_s305.complete.json	c4_timing_complete
lora_both_n500_s303.complete.json	c4_timing_complete
lora_both_n8000_s303.complete.json	c4_timing_complete
lora_dark_n2000_s303.complete.json	c4_timing_complete
lora_dark_n2000_s304.complete.json	c4_timing_complete
lora_dark_n2000_s305.complete.json	c4_timing_complete
lora_dark_n500_s303.complete.json	c4_timing_complete
lora_dark_n8000_s303.complete.json	c4_timing_complete
private_full.complete.json	c4_timing_complete
private_raw.complete.json	c4_timing_complete
private_warp_high.complete.json	c4_timing_complete
private_warp_low.complete.json	c4_timing_complete
leak_check_render-host.json	result summary

Receipt filename	Status / schema
cq_baseline_2026-09-08.json	result summary
cq_baseline_chesscog_2026-09-08.json	result summary
cq_baseline_cvchess_2026-09-08.json	result summary
crop_diag_2026-09-09.json	result summary
member_cvchess_full.tsv.json	result summary
member_cvchess_raw.tsv.json	result summary
member_cvchess_warp_high.tsv.json	result summary
member_cvchess_warp_low.tsv.json	result summary
member_slcctest_full.tsv.json	result summary
member_slcctest_raw.tsv.json	result summary
member_slcctest_warp_high.tsv.json	result summary
member_slcctest_warp_low.tsv.json	result summary
analysis.json	result summary
score_cq_cpu_20261004.json	result summary

F.1 Validation scope

The paper build checks the complete cell matrix, deterministic table generation, per-cell source comments, and rejection of altered evidence or broken historical hash links. The export regressions separately test token boundaries, paired original/export hashes, protected scene records, and explicit PDF decoder failures. Passing these checks does not validate the scientific claims in the paper text or resolve a distribution licence.

F.2 Conditional localization diagnostic

Figure 7 retains the earlier detector summary as a separate measurement from recognition. This figure and the historical pipeline comparison use hash-pinned aggregate summaries without original run-completion receipts; copying their hashes does not retrospectively certify the original runs. Error is normalized by the mean projected board edge, and the figure reports excluded detections beside the elevation bands.

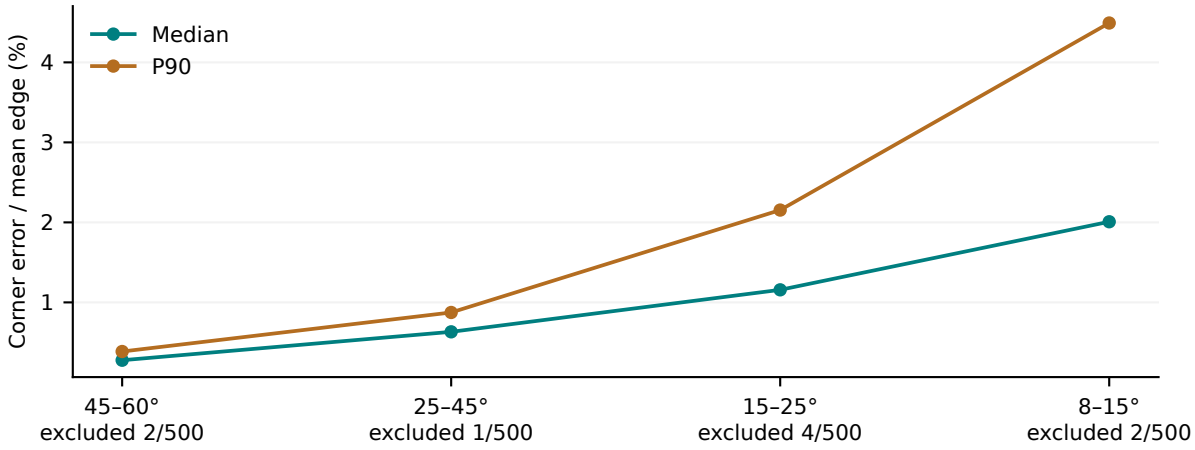


Figure 7: Conditional corner error normalized by mean projected board edge. Excluded detections are reported for each elevation band. Historical aggregates are hash-pinned; original run-completion receipts are unavailable.